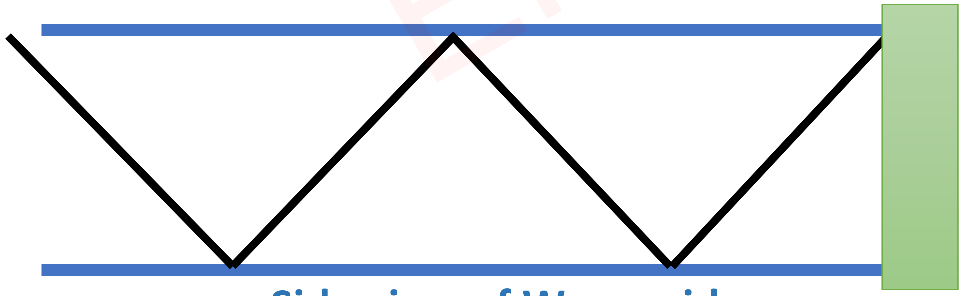
Waveguide Discontinuities

Why to use Waveguide Discontinuities?

- In Multiport Network, even if we have an ideal waveguide, there can be a loss of EM wave and reflection of the signal at the port.
- To minimize power loss and reflection of the signal, we use Waveguide Discontinuities.

Basics of Waveguide Discontinuities

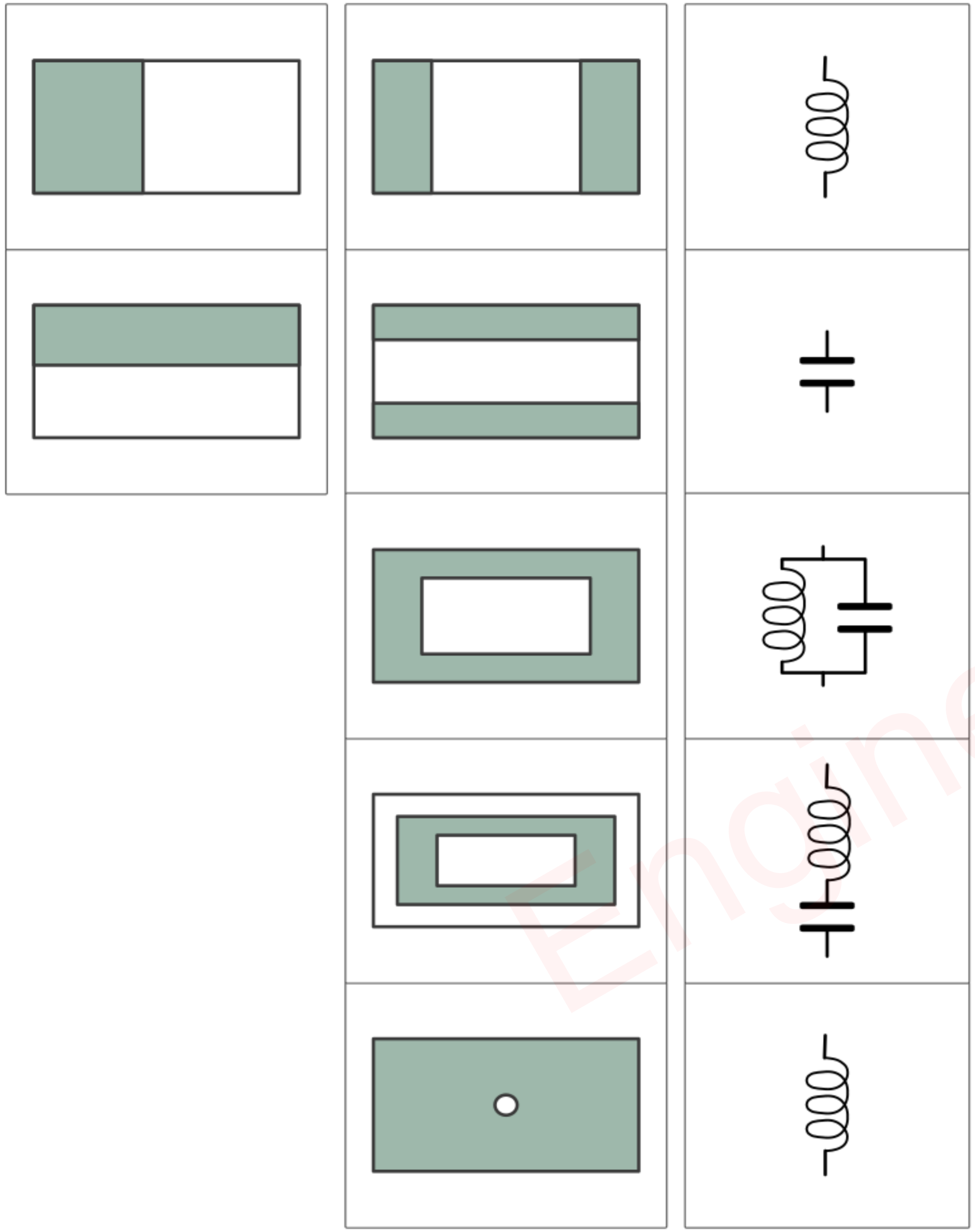
- Waveguide discontinuities are interruptions or changes in the structure of a waveguide that affect the propagation of electromagnetic waves.
- Issues in EM wave Propagation can occur due to the following reasons:
  - Changes in Waveguide Cross Section
  - Introduction of Obstacles
  - Variation in Dielectric Properties
  - Impedance Mismatch
- To avoid Power Loss and Impedance Mismatch we use the following methods of Waveguide Discontinuities:
  - Waveguide Irises
  - Tuning Screws and Posts
  - Matched Termination
- In Waveguide Discontinuities, we add lumped Parameters {Inductor, Capacitor}.

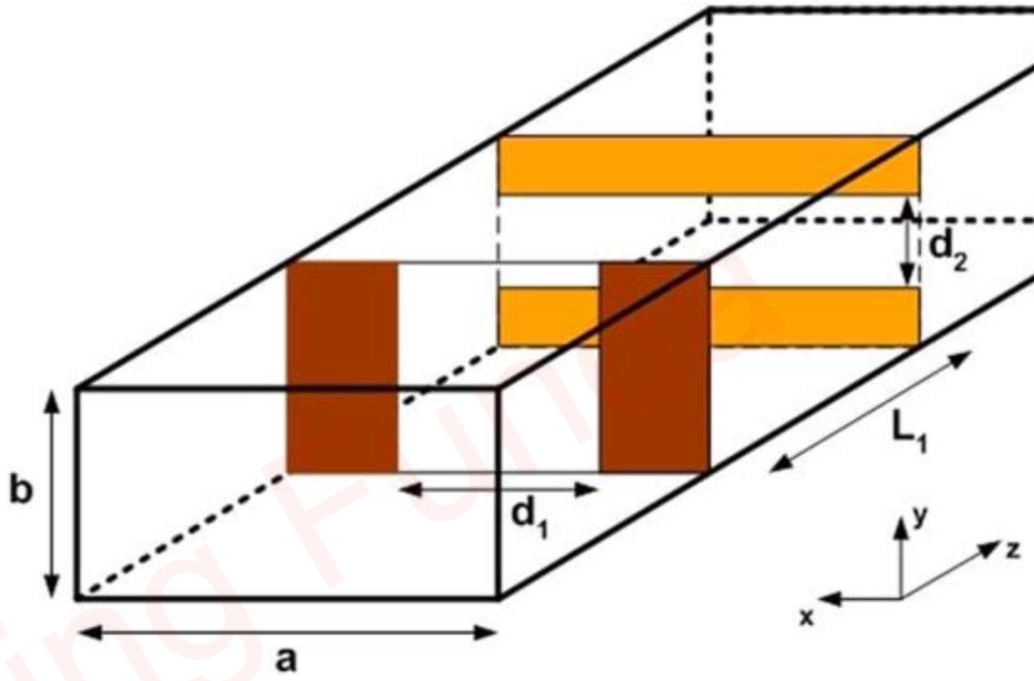


Waveguide Discontinuities to avoid reflection of EM waves.

Side view of Waveguide

Waveguide Irises

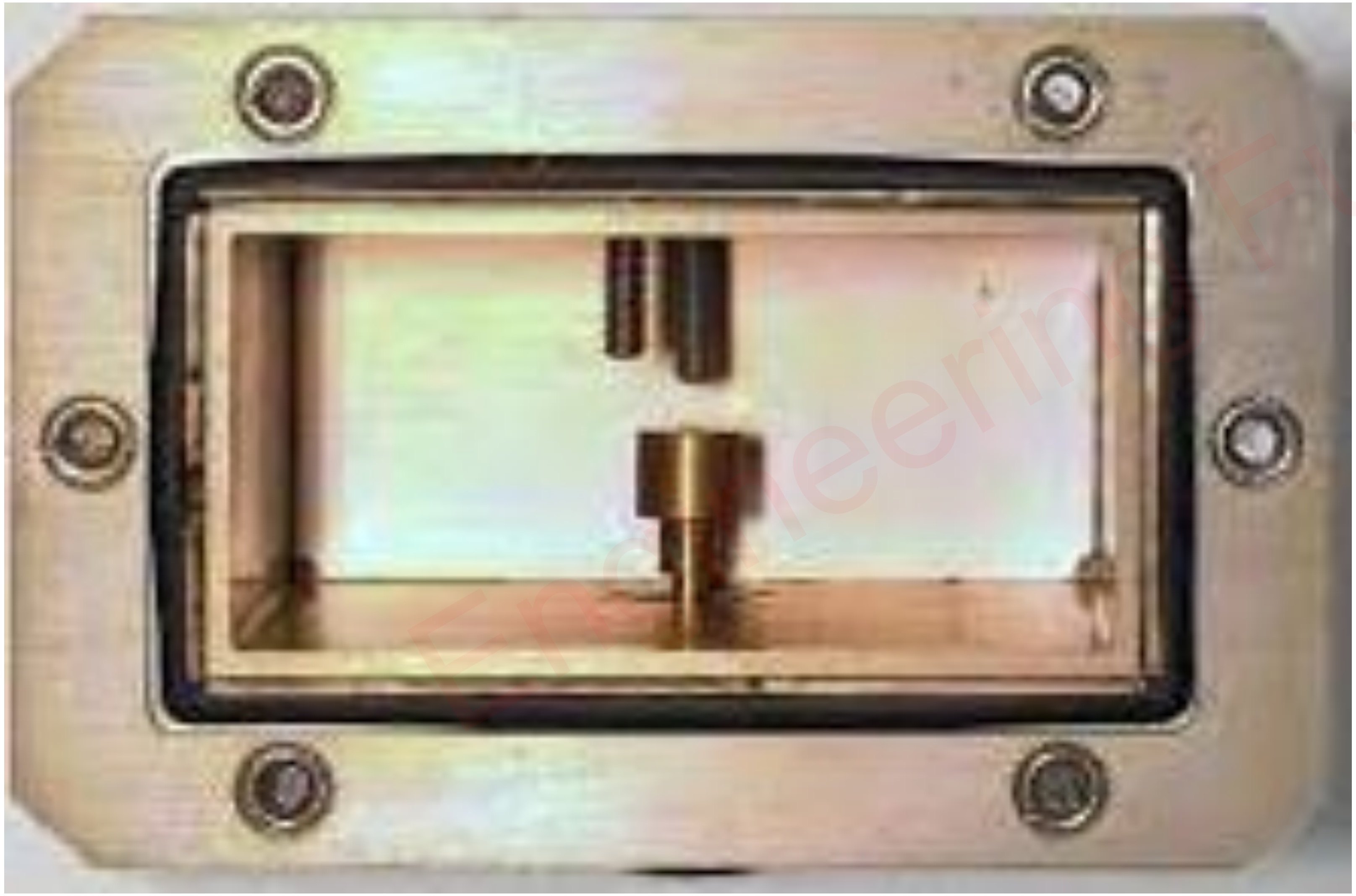




- ❑ Waveguide irises are window-like structures formed by a Diaphragm.
- ❑ Waveguide irises are placed on the Waveguide's Top, Bottom, or Sidewalls.
- ❑ Waveguide irises can control the strong Electric and Magnetic fields.

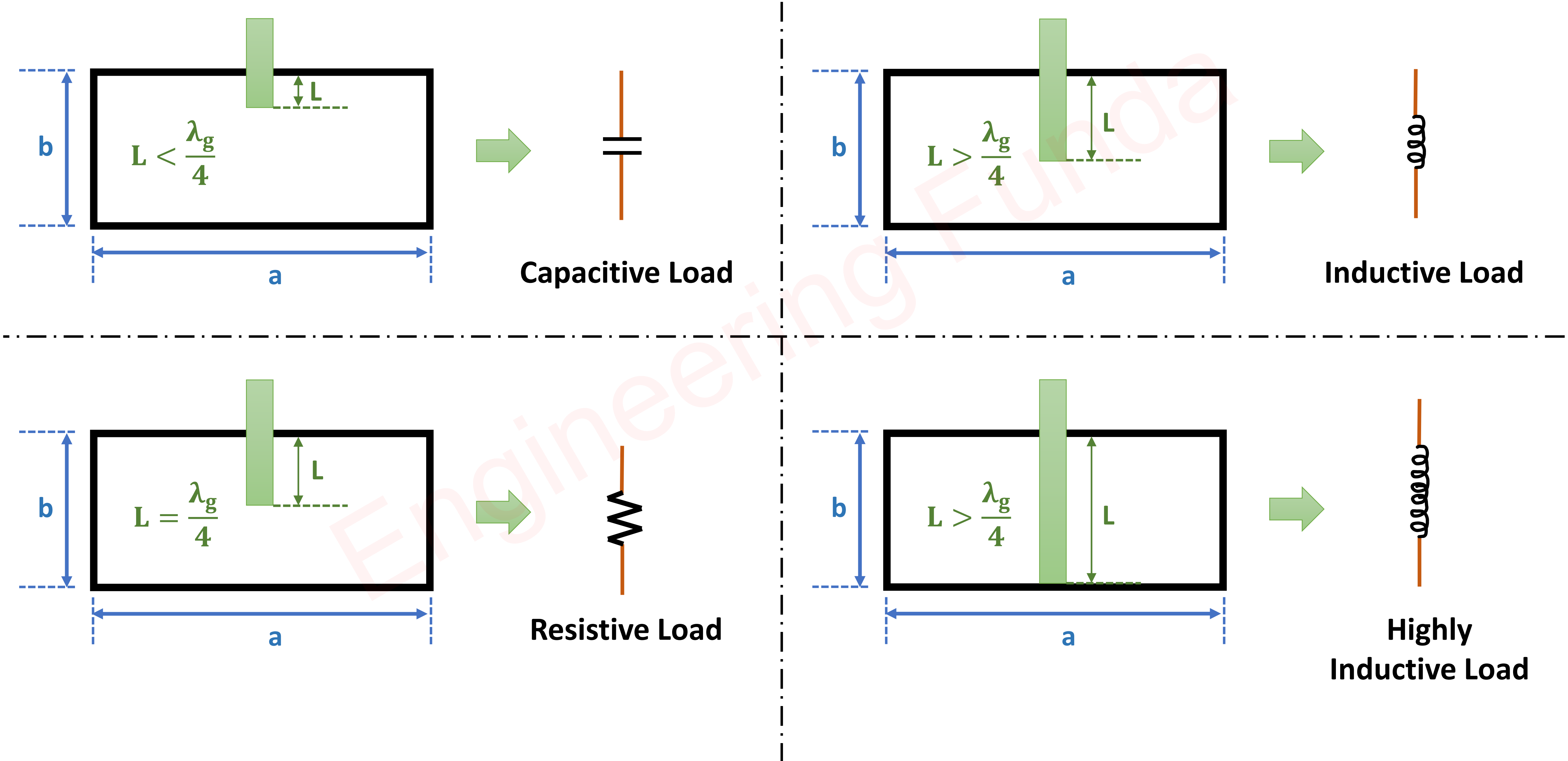
Tuning Screws and Posts

- Post is a rod-type material, which is inserted into a waveguide to absorb the EM wave.
- The Depth of insertion decides the type of the load.

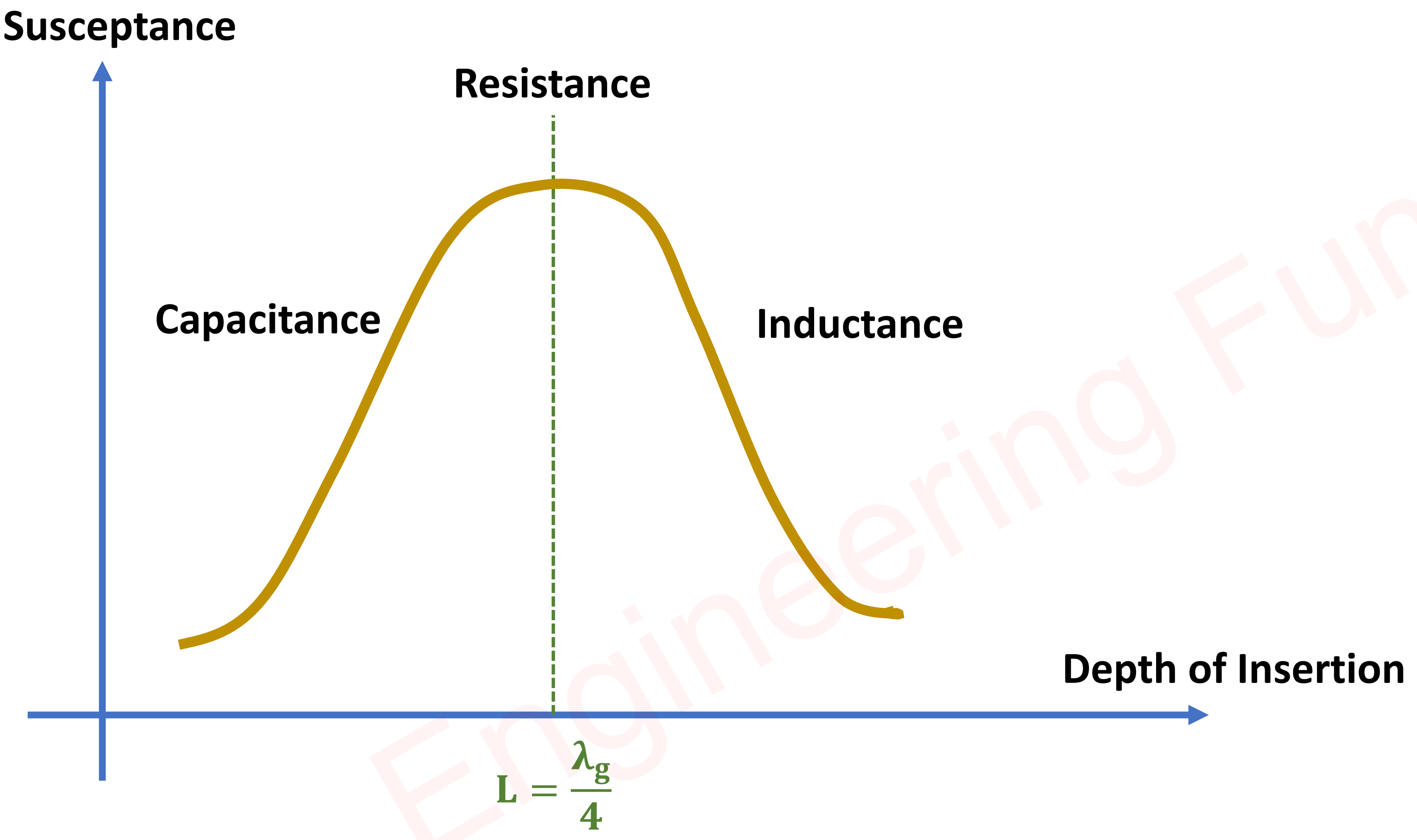


Equivalent Models of Tuning Screws and Posts

❑ The Depth of insertion decides the type of the load.

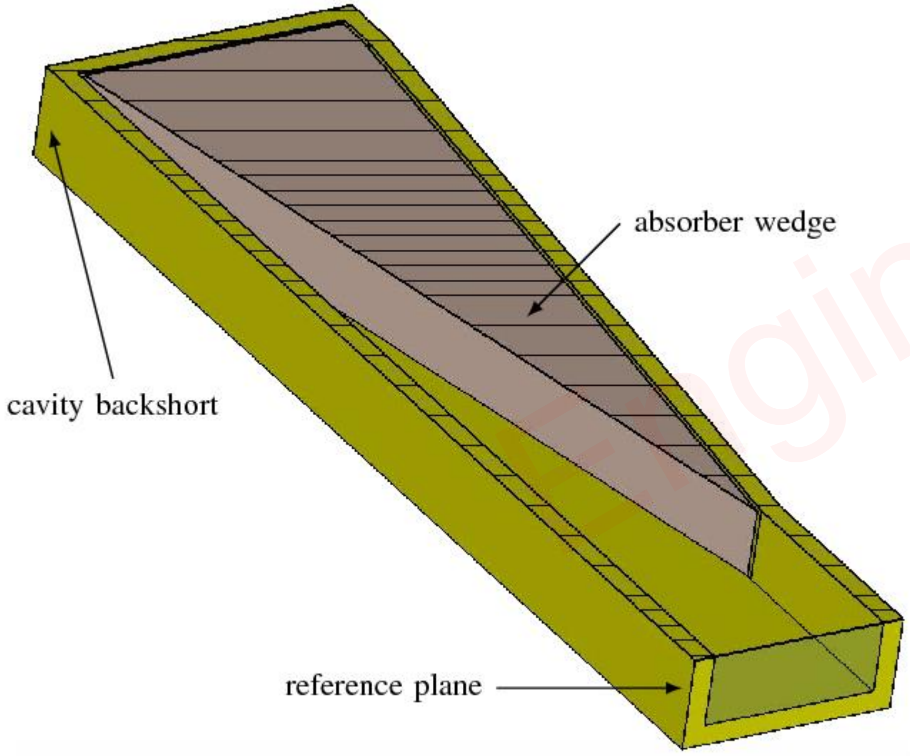


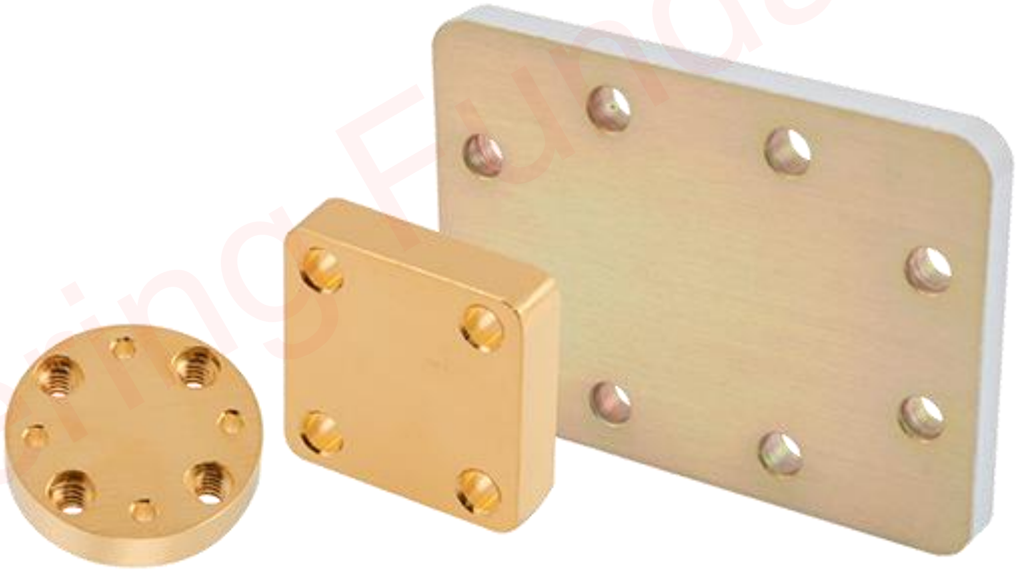
Relation between depth of insertion and Load



Matched Termination

- ❑ It provides perfect Impedance matching at ports.
- ❑ It does not allow any power reflection.
- ❑ There are three types of matched termination in general:
  1. Wedge Type Resistor {Attenuator}
  2. Plate Type
  3. Plunger Type

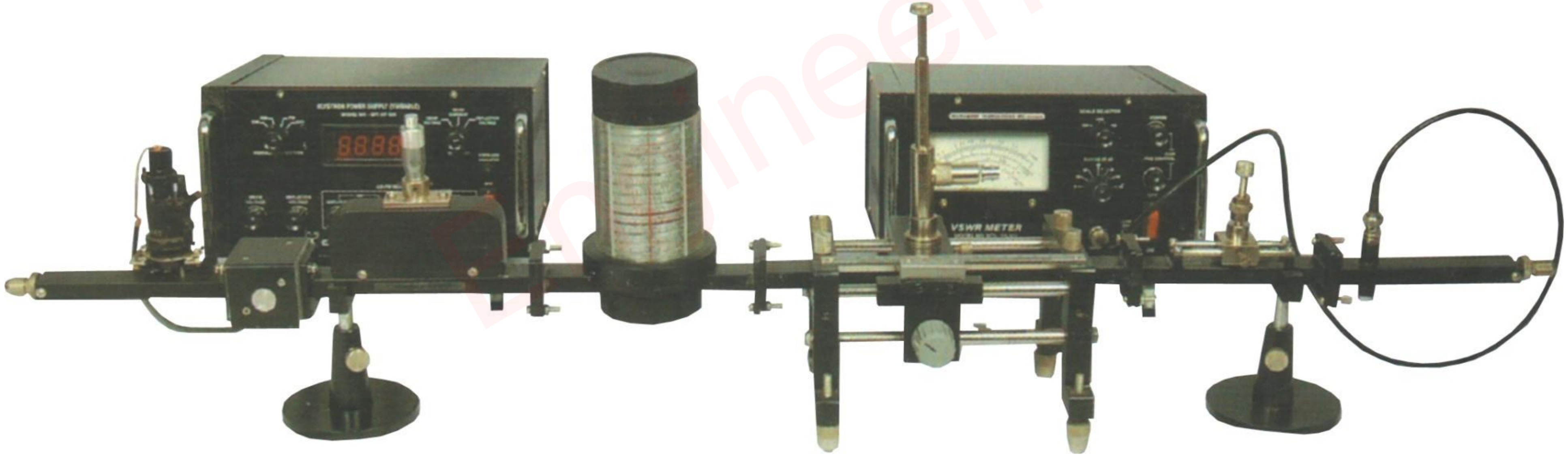




Coupling Mechanisms in WGs

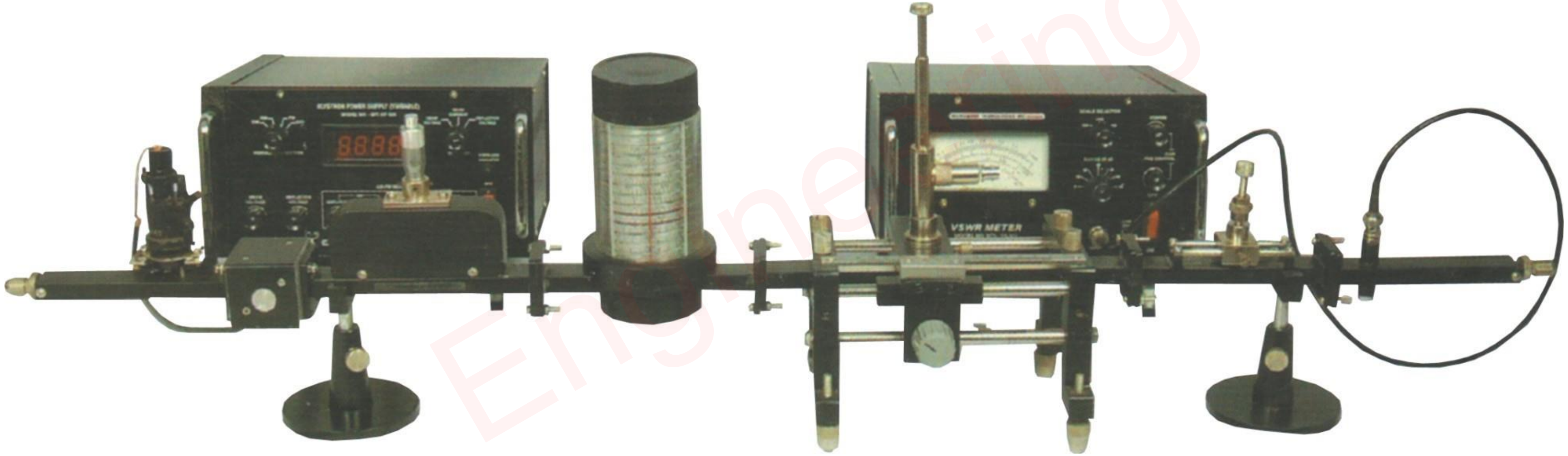
Basics of Coupling in Waveguides

- Waveguide Coupling is a process of transferring electromagnetic {EM} energy from one waveguide to another waveguide.
- Waveguides can be coupled to generators, excitation sources, or other waveguides to exchange energy in and out.
- During waveguide coupling, various propagation modes are excited, and there can be evanescent modes.



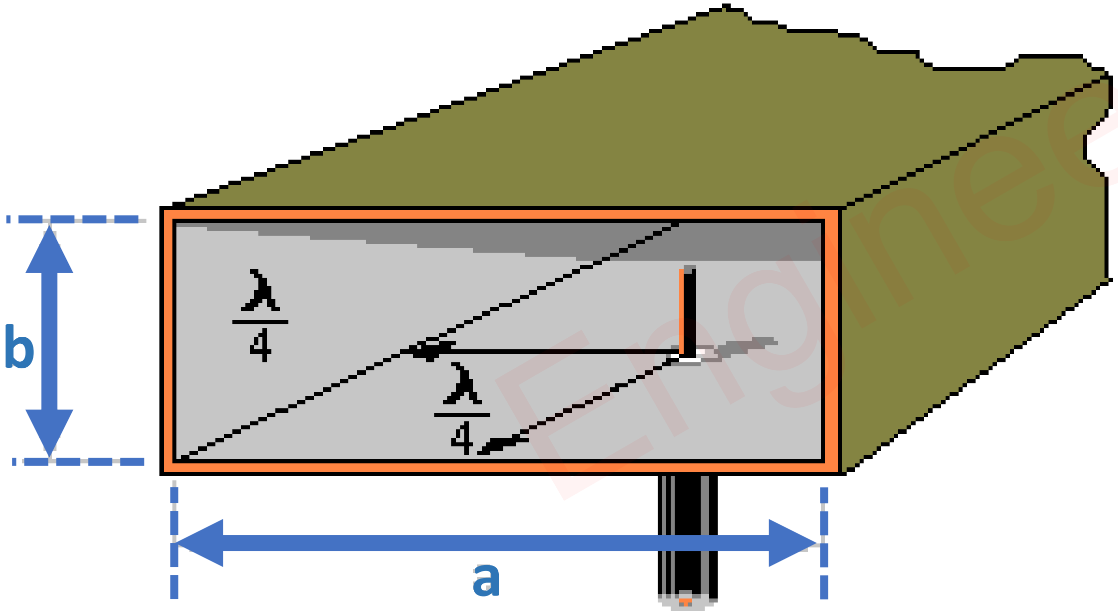
Types of Waveguide Coupling

- In General, there are three types of waveguide coupling mechanisms:
  - Probe Coupling – In Probe Coupling, Only the Electric Field can be coupled in Waveguide. Here, Magnetic Energy is not transferred in the waveguide.
  - Loop Coupling – In Loop Coupling, Only the Magnetic field can be coupled in waveguide. Here, Electric Energy is not transferred in the waveguide.
  - Aperture Coupling – In Aperture Coupling, EM waves are transferred from one waveguide to another waveguide.

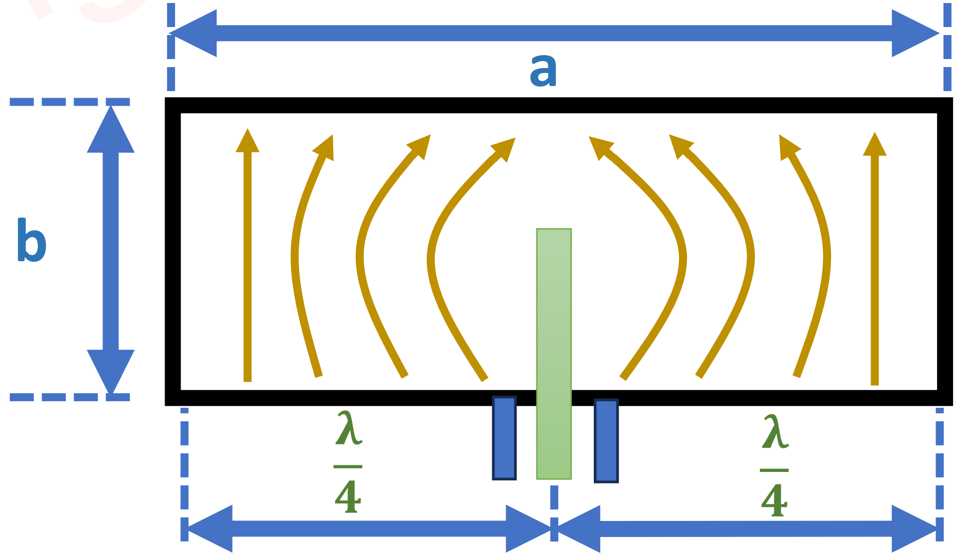


Probe Coupling in Waveguide

- Coaxial Line used in Probe Coupling.
- The Inner Pin of the Coaxial Line is inserted through a small hole at the location of $\lambda/4$ as shown in the figure.
- Using Probe Coupling, only Electric Field Energy is released in Waveguide.
- The Probe radiates equal Electric Energy in the waveguide as shown in the figure.
- The Probe (Pin) (Antenna) is inserted at the location perpendicular the length of the waveguide.



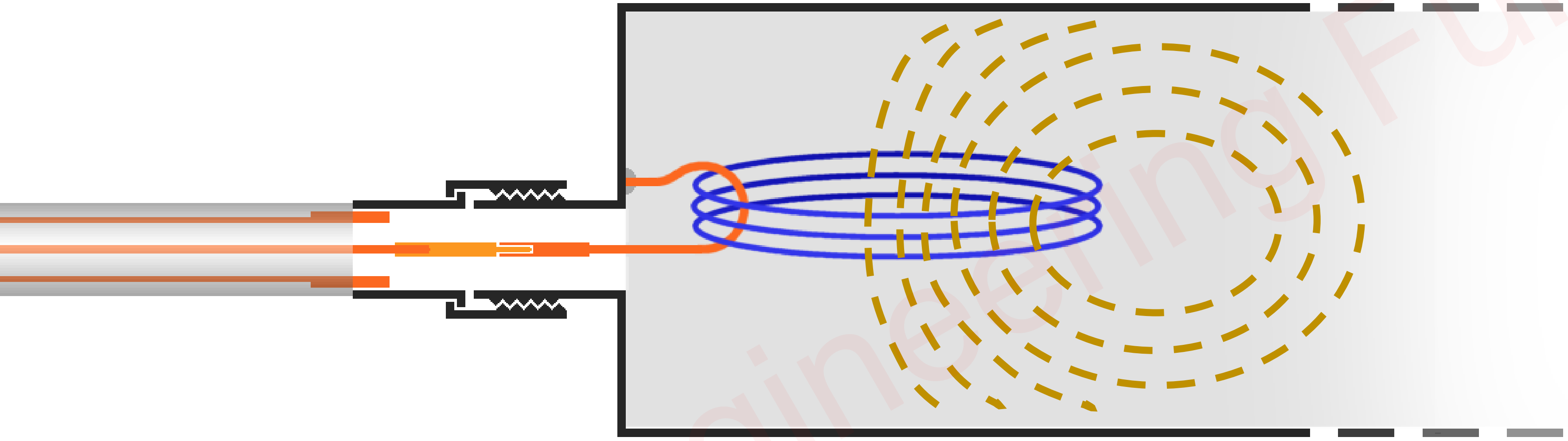
Coaxial Cable



Coaxial Cable

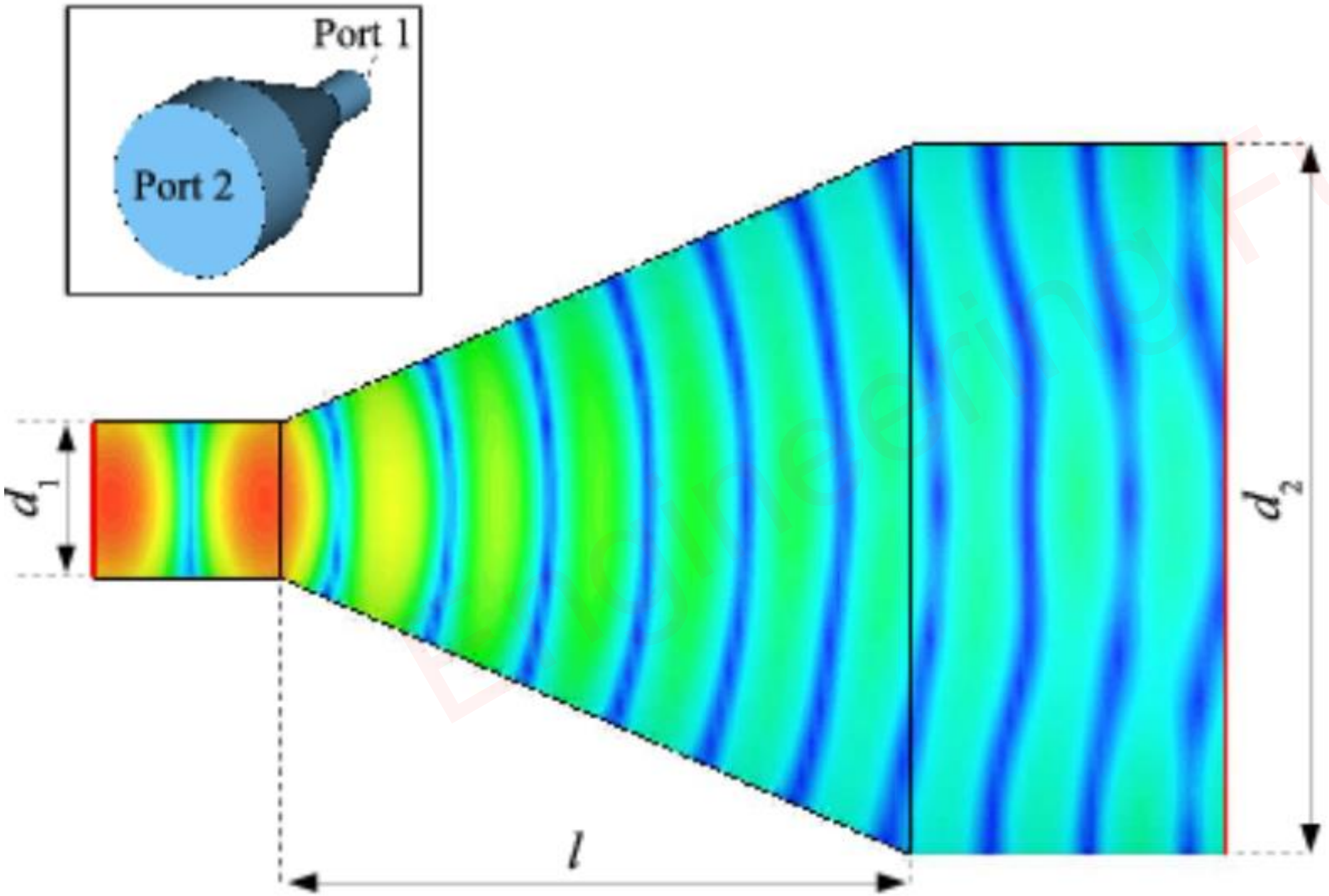
Loop Coupling in Waveguide

- Coaxial Line used in Loop Coupling.
- The Inner Pin of the Coaxial Line is inserted through a small hole and bands into the loop inside the waveguide.
- Using Loop Coupling, only Magnetic Field Energy is released in Waveguide.



Aperture Coupling in Waveguide

- It is used to couple Electric field and Magnetic Field Energy from one waveguide to another waveguide through the Aperture of the waveguide.



Waveguide Attenuators

Basics of Waveguide Attenuator

- Waveguide Attenuator is an electronic device that reduces the power of a signal without appreciably distorting its waveforms.

Input
15dB

Output
15dB - Attenuation

Variable Attenuation

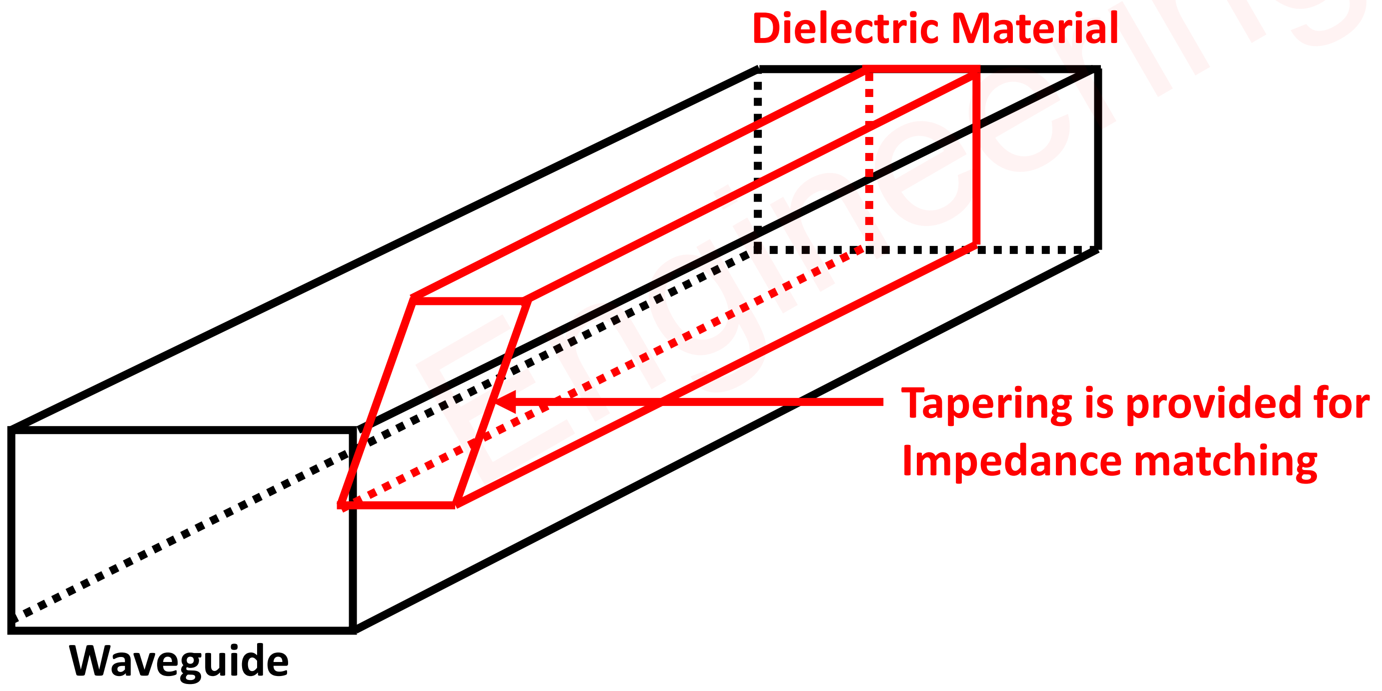
Input
15dB

Output
15dB – 6dB = 9dB

Fixed Attenuation {6dB}

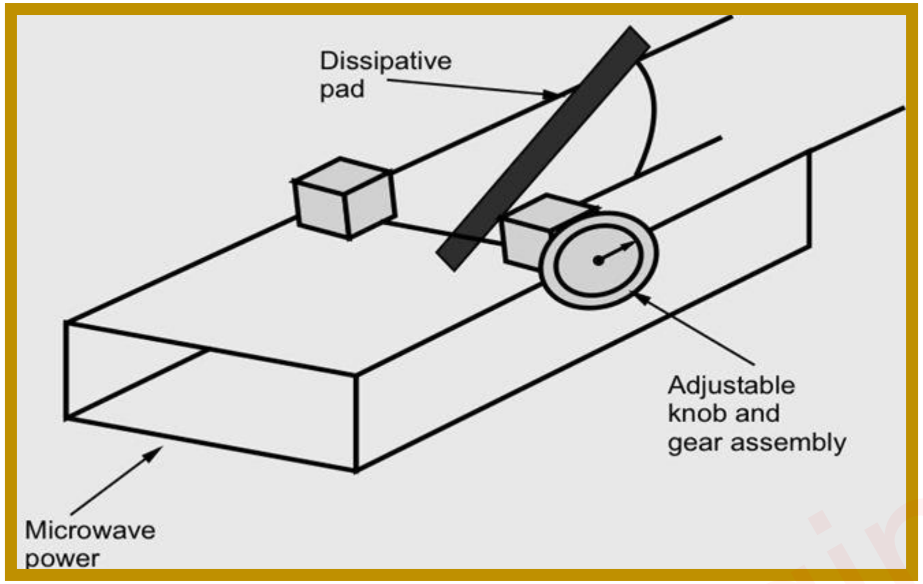
Fixed Attenuators

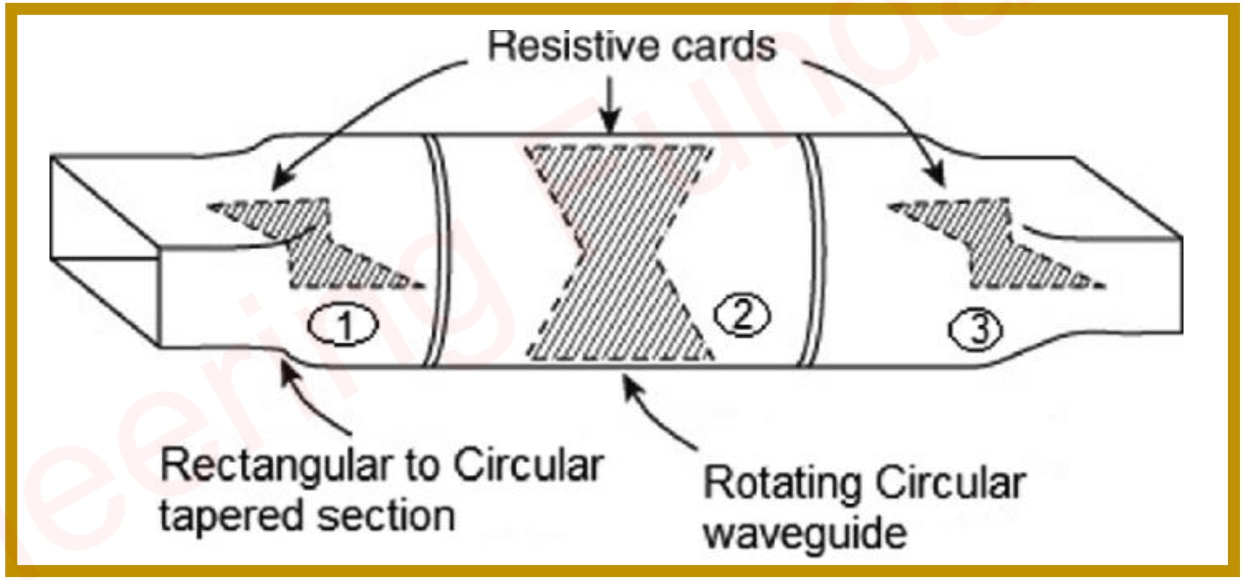
- Fixed Attenuators in circuits lower voltage, dissipate power and improve impedance matching.
- Fixed Attenuator consists of a dielectric slab which is a glass slab coated with thin carbon film to absorb the EM power.
- Fixed Attenuators are commonly used where a fixed amount of power is to be provided.
- If the Entire power enters in the waveguide is Nuffield then that Attenuator is called Terminator.

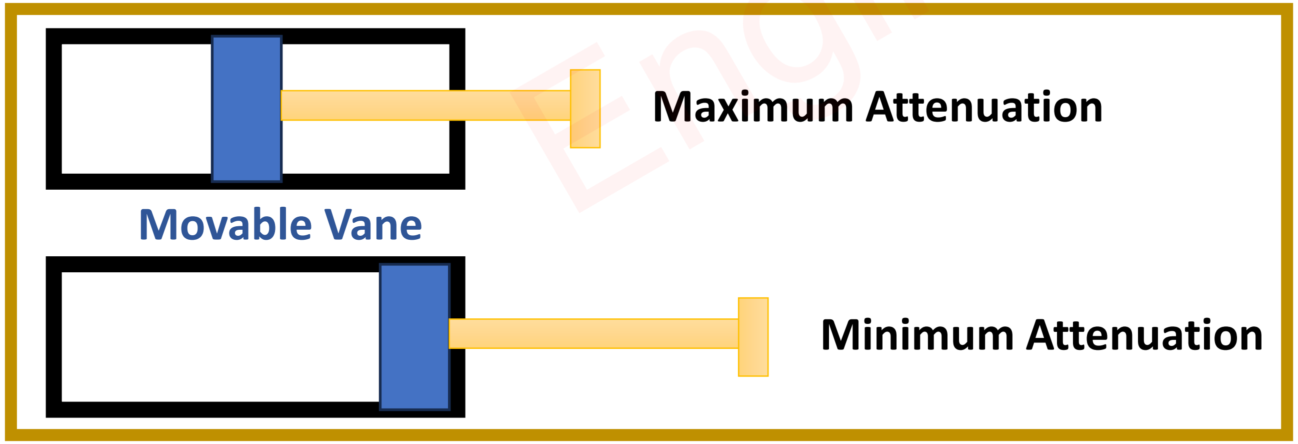


Variable Attenuators

- Variable Attenuators provide continuous or stepwise variable attenuation.
- For Rectangular Waveguides, Variable Attenuators can be of Flap Type or Vane Type.
- For Circular Waveguides, Variable Attenuators can be of Rotary Type.



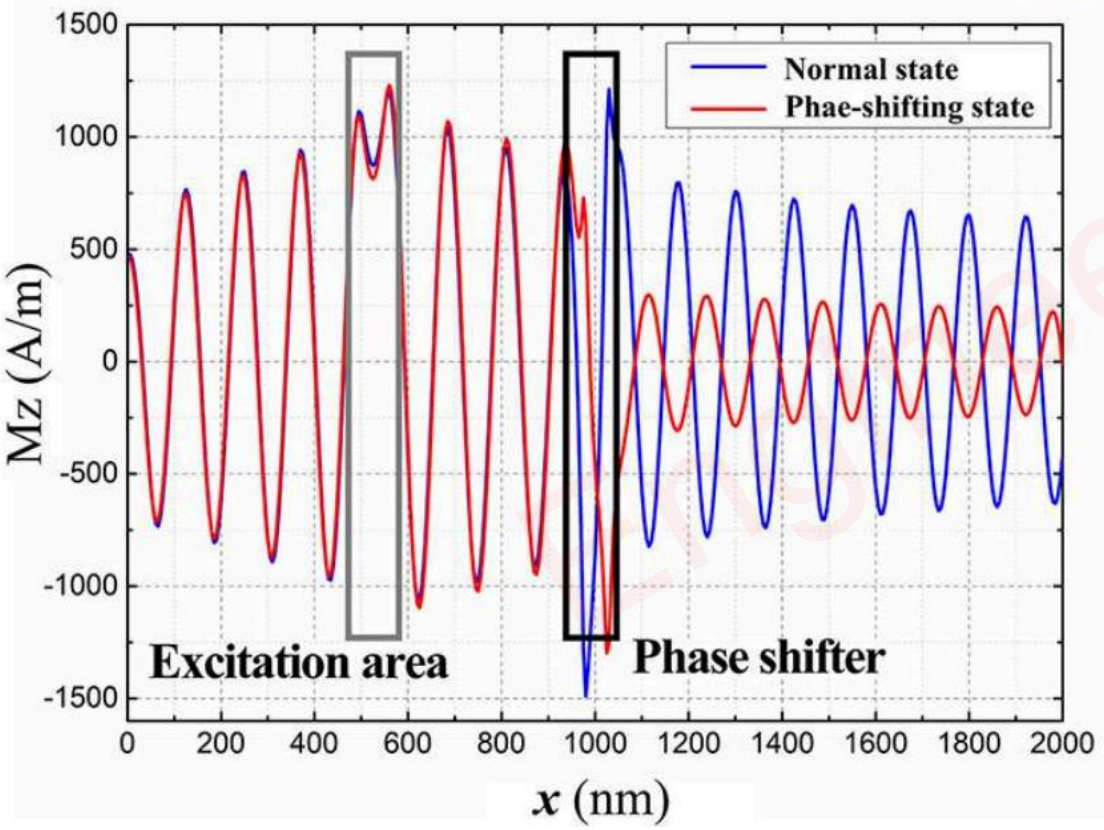


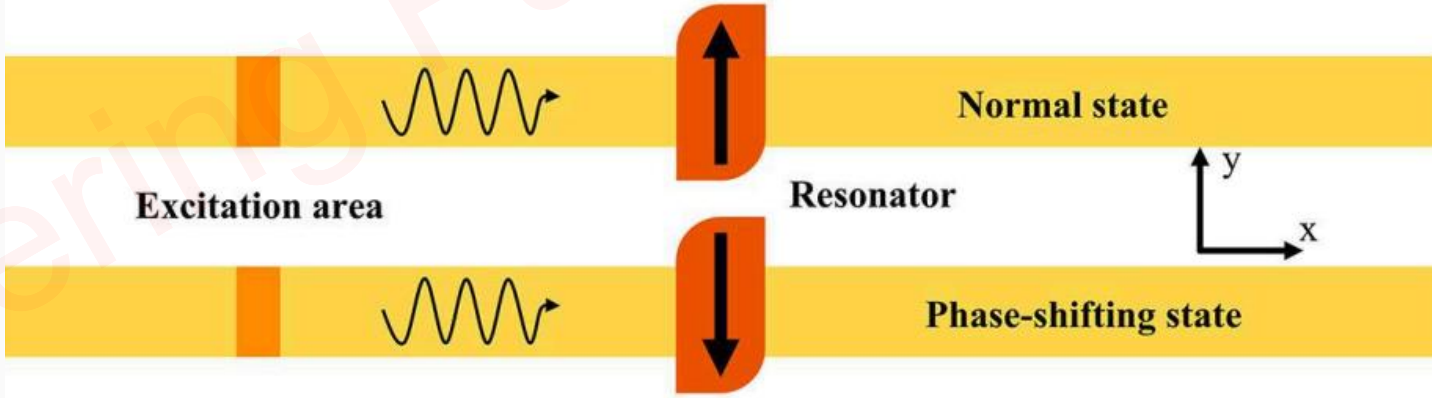


Waveguide Phase Shifter

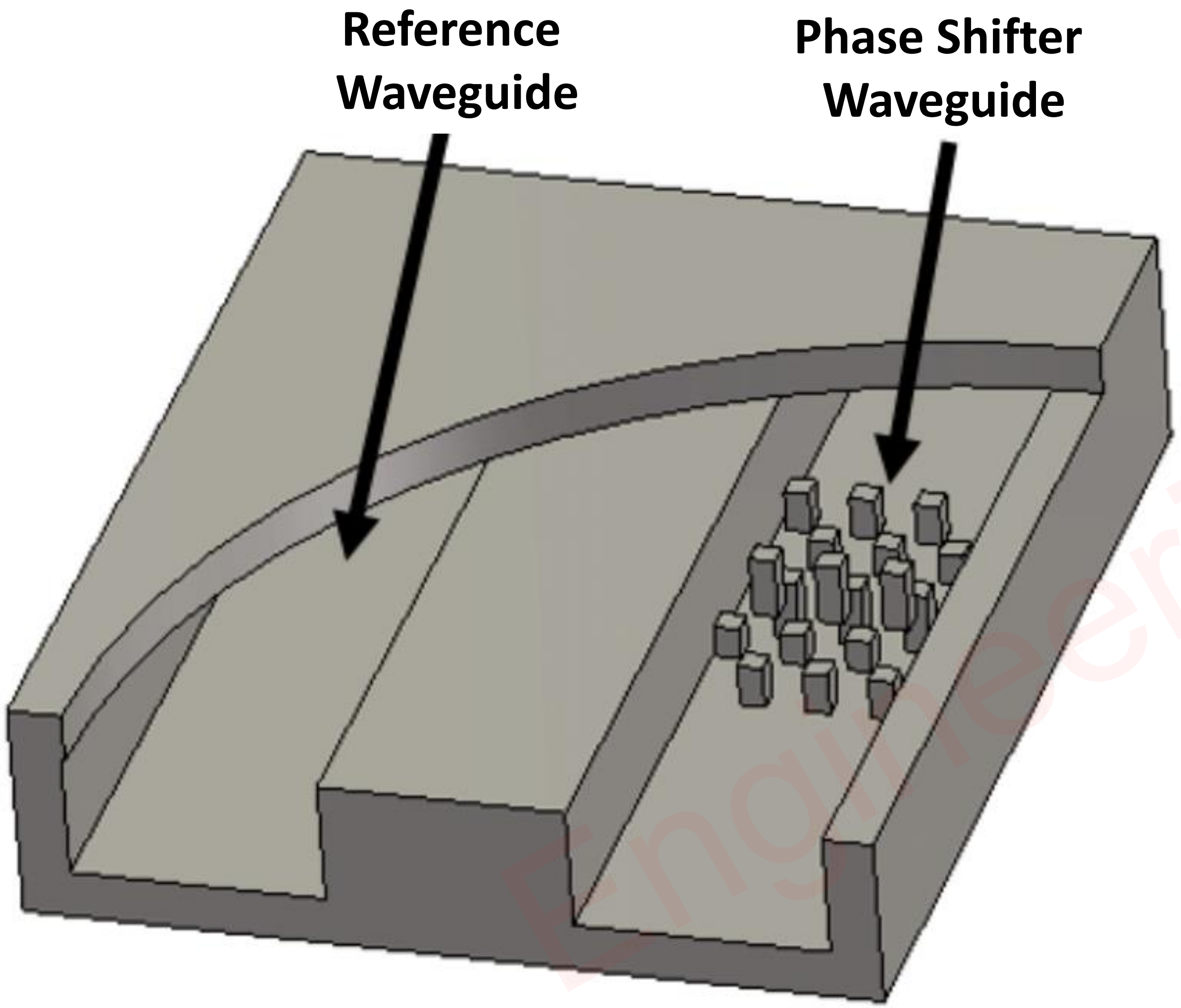
Basics of Waveguide Phase Shifter

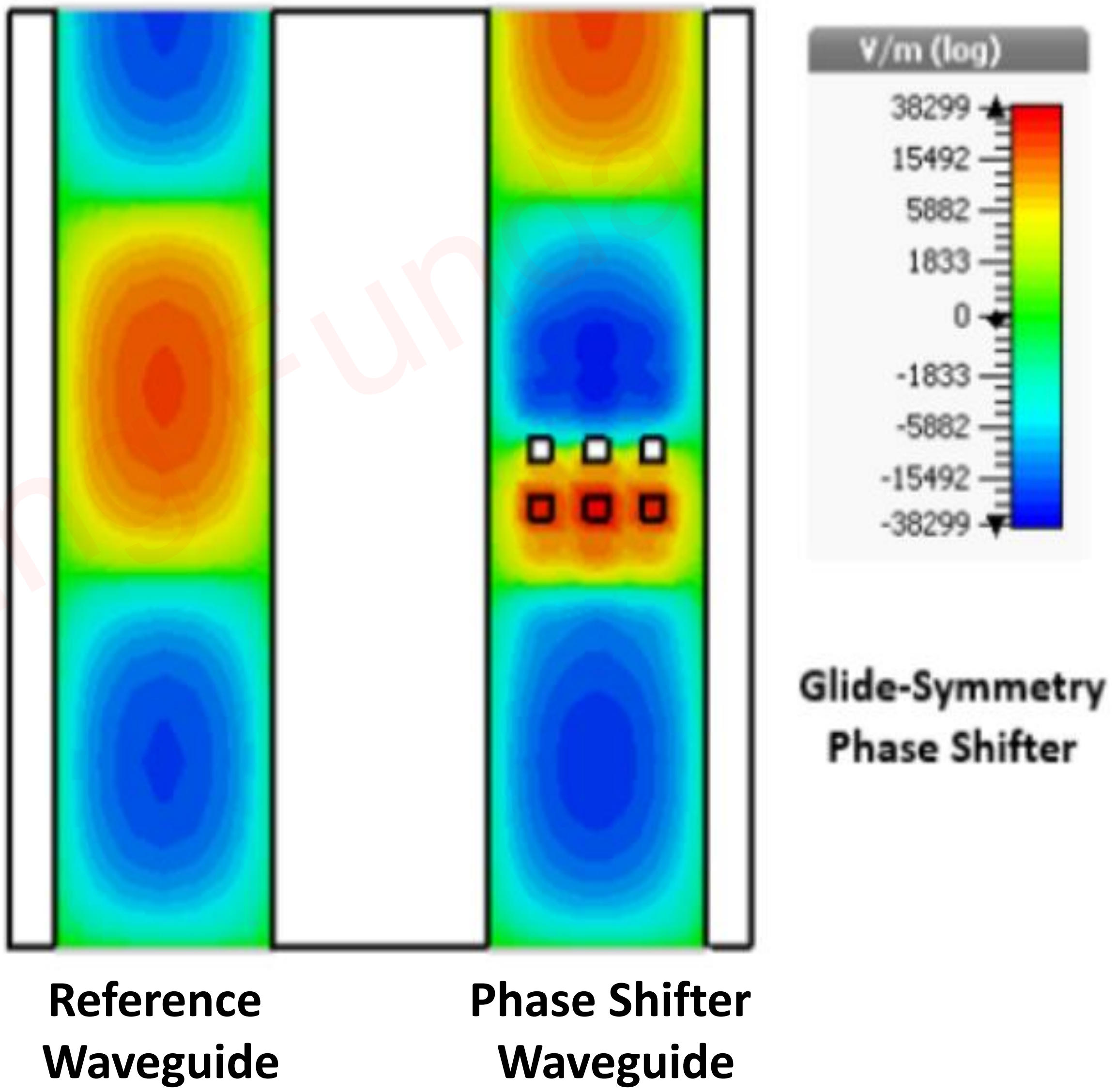
- Waveguide Phase Shifter is used to provide Phase Shift in Input Signal.
- Waveguide Phase Shifter is a two-port device, That is used to provide phase shift in RF-signal with practically negligible attenuation.



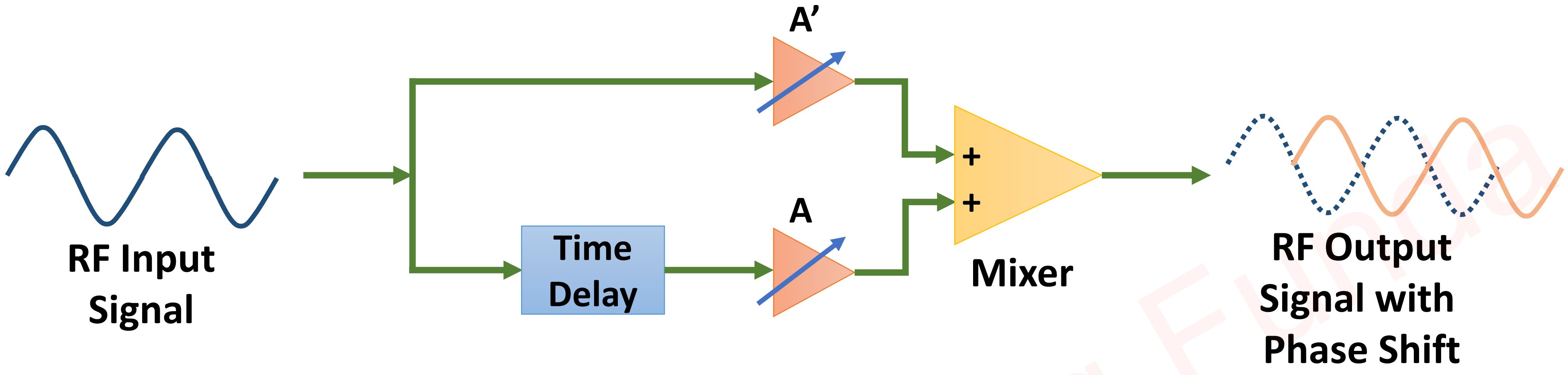


Glide Symmetric Waveguide Phase Shifter





Block Diagram of Waveguide Phase Shifter



- ❑ Waveguide Phase Shifter is used to provide Phase Shift in Input Signal.
- ❑ Waveguide Phase Shifter is a two-port device, That is used to provide phase shift in RF-signal with practically negligible attenuation.
- ❑ The Phase Shift depends on waveguide length and Operating wavelength.
- ❑ There are two types of Phase Shifters:
  - Adjustable Phase Shifter
  - Non-Adjustable Phase Shifter

Adjustable Phase Shifter

- It consists of a feeder section that introduces a phase shift at a certain frequency.
- The Magnitude of phase shift can be adjusted as required.

Non-Adjustable Phase Shifter

- It consists of a phased calibrated feeder section.
- Phase shift is provided by selecting the desired length of the feeder section, Cross-sectional dimensions, and Effective Dielectric Constant.

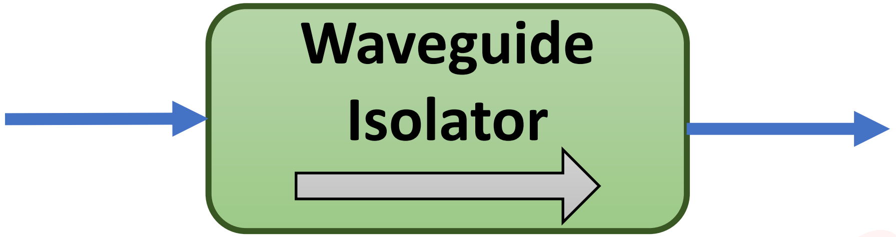
Applications of Phase Shifter

- RADARs and Communication Systems
- Smart Antennas
- Antenna Arrays
- Microwave Instrumentation & Measurement System

Waveguide Isolator

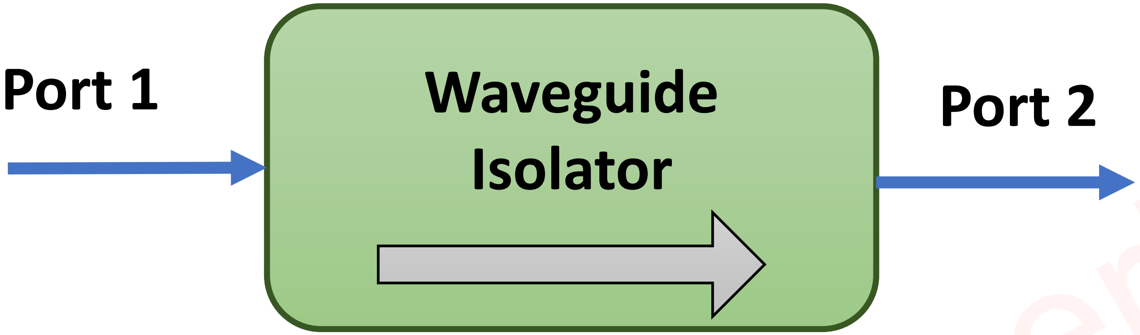
Basics of Waveguide Isolator

- Waveguide Isolator is a Two-Port device.
- Waveguide Isolator is used to Isolate One port from Another port.
- In Waveguide Isolator, Signal flows in only one direction.
- Waveguide Isolator used just after the Microwave source in the Microwave test bench.



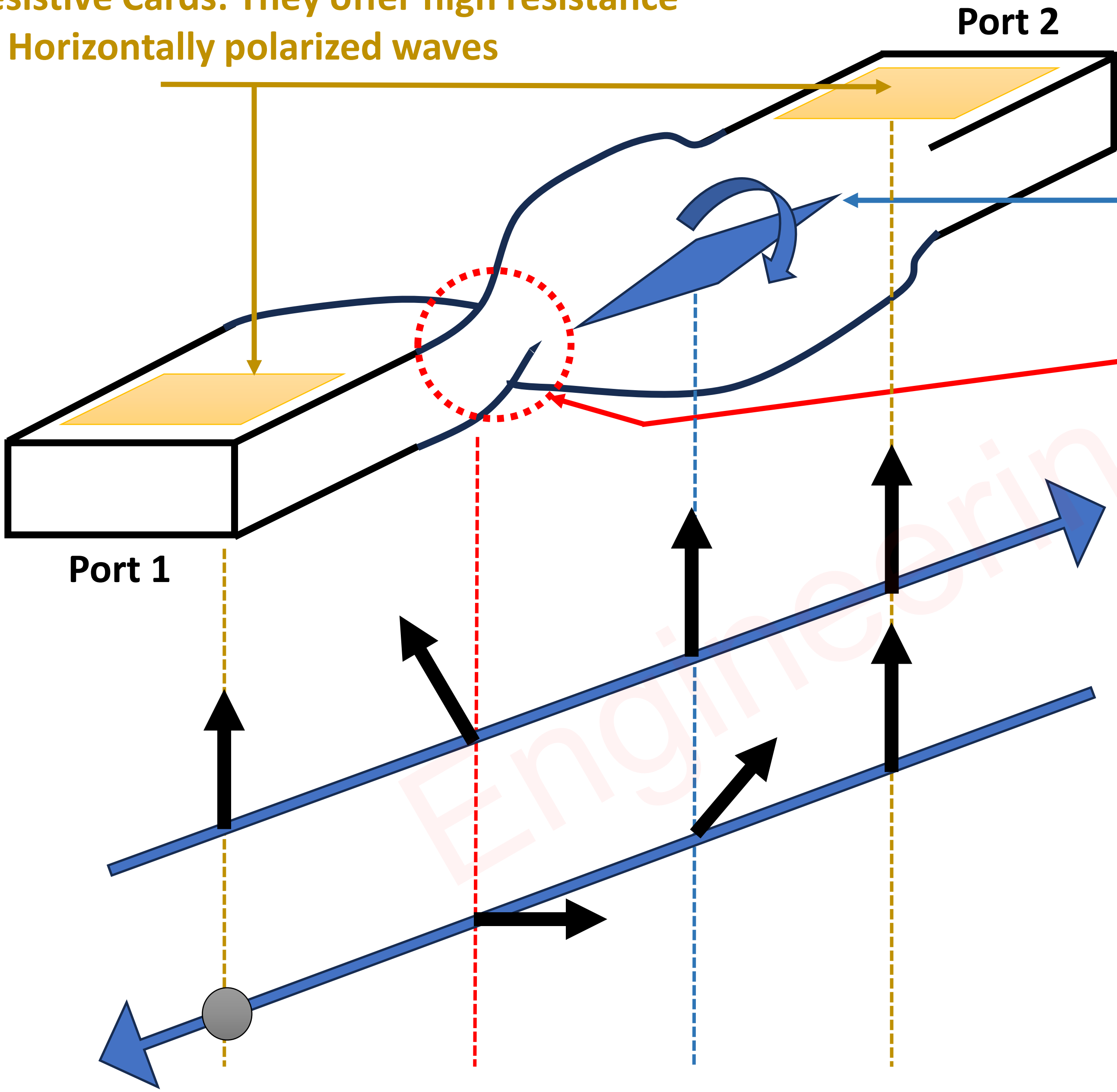
Working of Waveguide Isolator

- ❑ If The Input is given at port 1, The wave will propagate From port 1 to port 2 without any Attenuation.
- ❑ If The Input is given at port 2, The wave will not propagate From port 2 to port 1. It produces infinite Attenuation. Hence Output port 1 will be zero.



Internal Structure of Waveguide Isolator

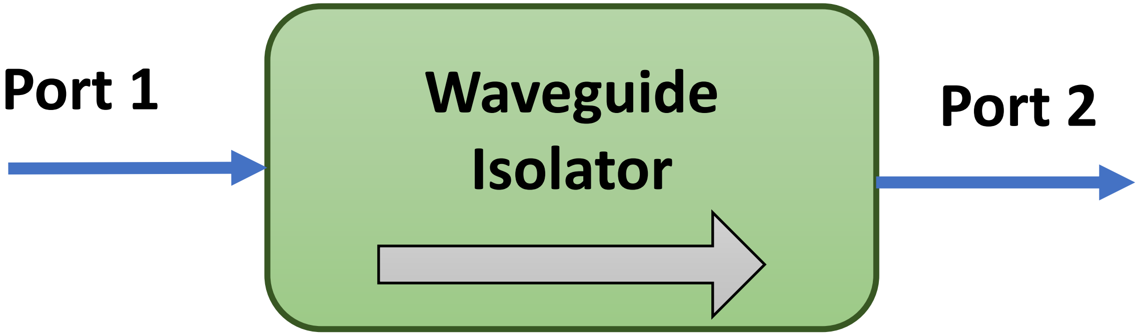
Resistive Cards: They offer high resistance
to Horizontally polarized waves



Ferrite Rod: It Rotates E Field by 45 Degrees.

45^0 Twister: It Rotates E Field by 45 Degrees
in Anticlockwise direction.

S Matrix of Waveguide Isolator



- The Isolator is a Two-Port Device.
- So, It's S-Matrix is (2 x 2) Matrix.

$$[S] = \begin{vmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{vmatrix}$$

- If port 1 and port 2 are perfectly matched, then $S_{11} = S_{22} = 0$.
- If Input is given at port 2, Output at port 1 will be Zero, $S_{12} = 0$.

$$[S] = \begin{vmatrix} 0 & 0 \\ S_{21} & 0 \end{vmatrix}$$

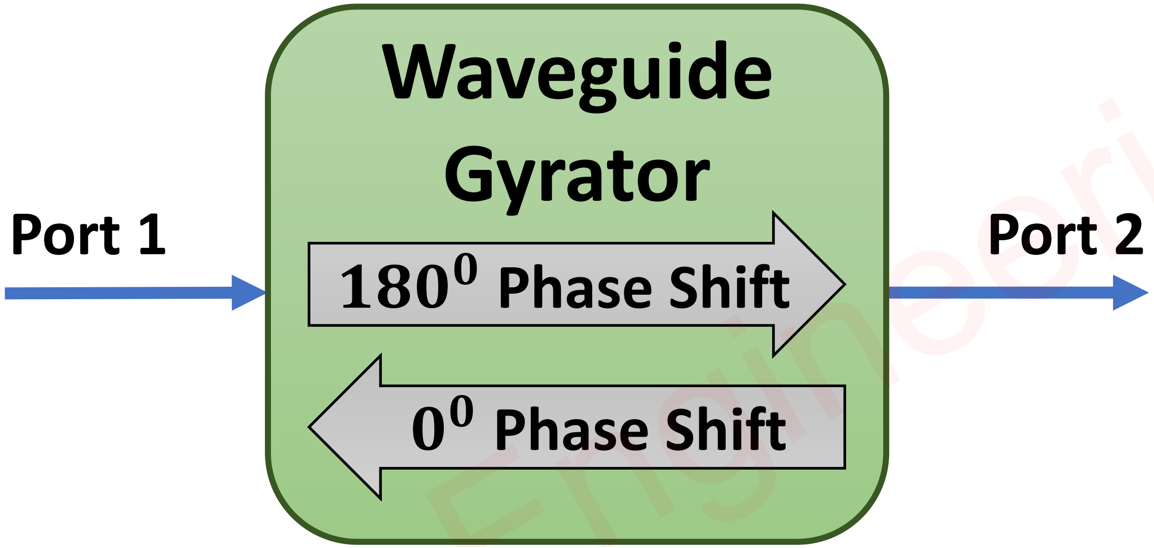
- Using Identity property, $[S][S]^* = [I] \Rightarrow \begin{vmatrix} 0 & 0 \\ S_{12} & 0 \end{vmatrix} \begin{vmatrix} 0 & 0 \\ S_{12}{}^* & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \Rightarrow S_{12} = 1$

$$[S] = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix}$$

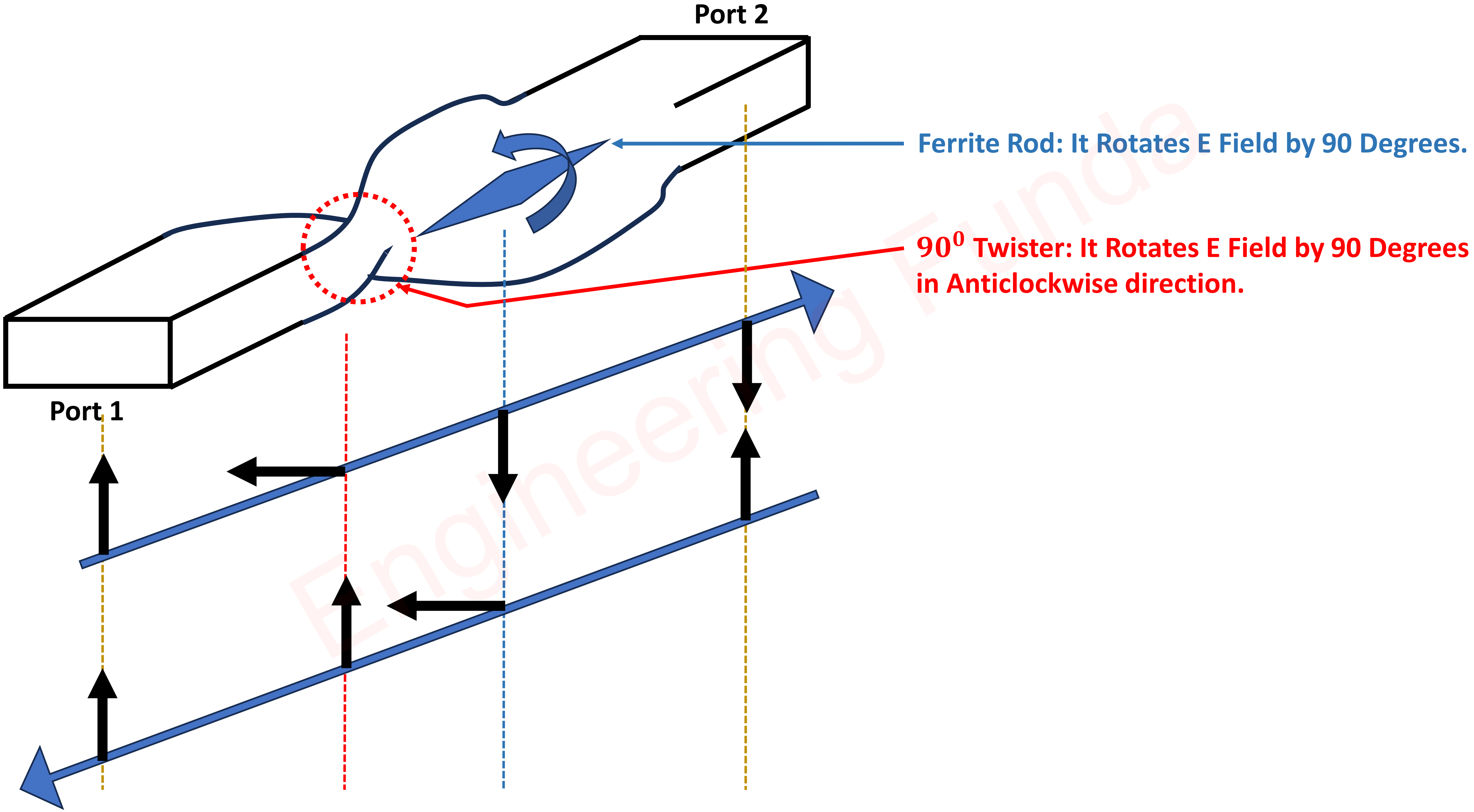
Waveguide Gyrator

Basics of Gyrator

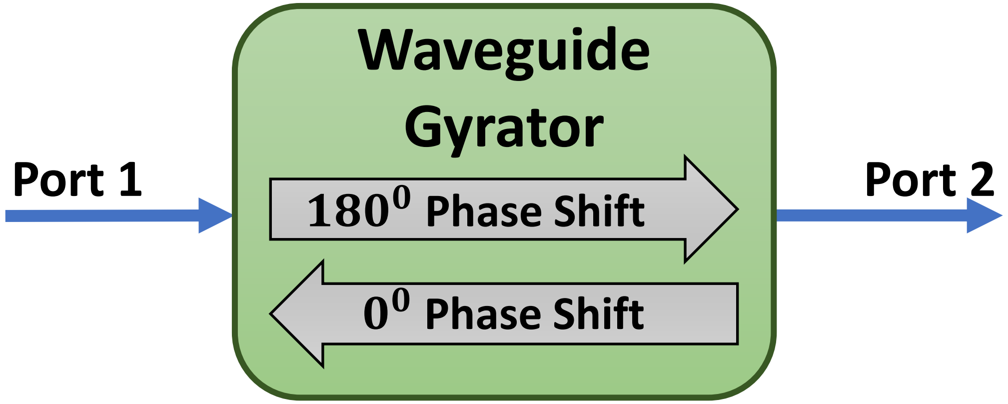
- Waveguide Gyrator is a Two-Port device.
- Waveguide Gyrator Provides 180^0 phase shift when signals flow from port 1 to port 2 compared to 0^0 phase shift when signal flow from port 2 to port 1.



Internal Structure of Waveguide Gyrator



S Matrix of Waveguide Gyrator



❑ The Waveguide Gyrator is a Two-Port Device. So, It's S-Matrix is (2 x 2) Matrix.

$$[S] = \begin{vmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{vmatrix}$$

❑ If port 1 and port 2 are perfectly matched, then $S_{11} = S_{22} = 0$.

❑ From the working of the Gyrator, $S_{12} = -S_{21}$.

$$[S] = \begin{vmatrix} 0 & S_{12} \\ -S_{12} & 0 \end{vmatrix}$$

❑ Using Identity property, $[S][S]^{*} = [I] \Rightarrow \begin{vmatrix} 0 & S_{12} \\ -S_{12} & 0 \end{vmatrix} \begin{vmatrix} 0 & S_{12}^{*} \\ -S_{12}^{*} & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \Rightarrow S_{12} = 1$

$$[S] = \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix}$$

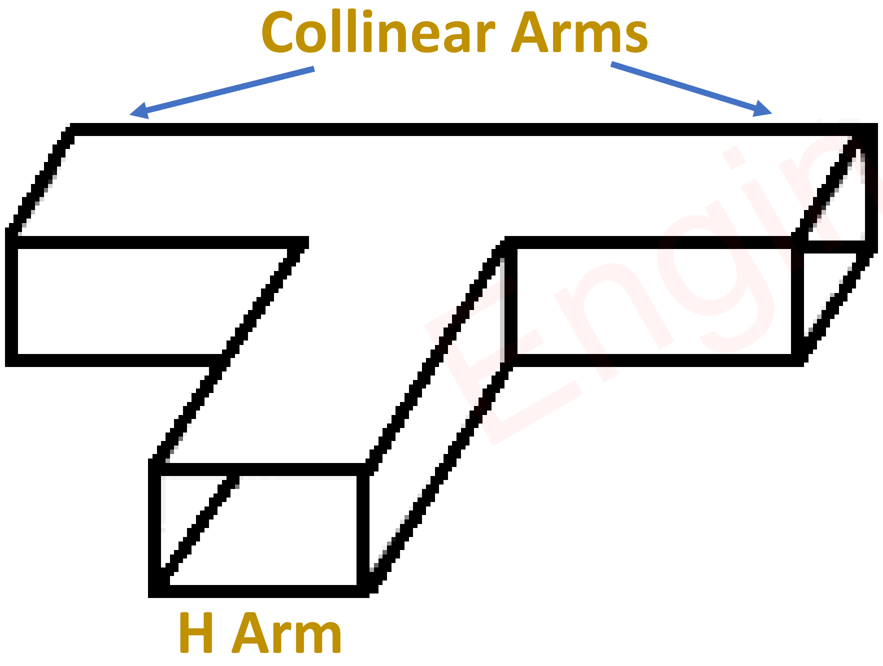
Applications of Gyrator

- The Primary goal of the Gyrator is to reduce the size of the circuit by removing the inductor.
- It is used in Telephony devices.
- Parametric Equalizer.
- Band Pass Filter.

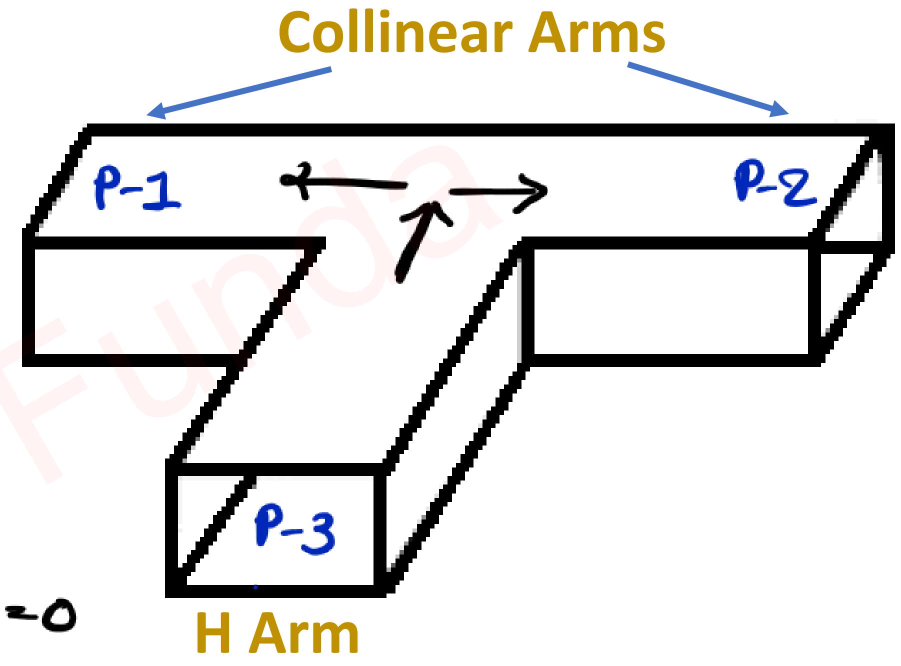
H – Plane Tee Junction

Basics of H – Plane Tee Junction

- H – Plane Tee Junction is a Three port device.
- H – Plane Tee Junction has one H Arm (Port) and Two Collinear Arms (Ports).
- Collinear Arms are Symmetric with respect to H Arm.
- H – Arm is perfectly matched arm.
- So, If Input is given to H-Arm, then Output at Collinear Ports is equal and in Phase.



Scattering Parameters of H – Plane Tee Junction



- H-plane Tee junction is 3-port device.

  So, S-Matrix is given by

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

$$S_{ij} = \frac{b}{a}$$

i^{th} o/p j^{th} i/p

- H-arm is perfectly matched Arm $\Rightarrow S_{33} = 0$
- If Input is given to H-Arm (Port 3), o/p at port-1 and port-2 is equal and In phase.

$$S_{13} = S_{23}$$

- As property Symmetry, $S_{ij} = S_{ji}$

$$S_{12} = S_{21}, \quad S_{13} = S_{31} = S_{23} = S_{32}$$

→ So Modified Scattering matrix will be

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix}$$

→ As per Identity property,

$$\Rightarrow [S][S]^* = [I]$$

$$\Rightarrow \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & S_{13}^* \\ S_{13}^* & S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ $R_3C_3 \Rightarrow S_{13}^2 + S_{13}^2 = 1 \Rightarrow S_{13} = 1/\sqrt{2}$

$$\left.\begin{array}{l} \rightarrow R_1C_1 \Rightarrow S_{11}^2 + S_{12}^2 + S_{13}^2 = 1 \Rightarrow S_{11}^2 + S_{12}^2 = 1/2 \quad \text{— (1)} \\ \rightarrow R_2C_2 \Rightarrow S_{12}^2 + S_{22}^2 + S_{13}^2 = 1 \Rightarrow S_{12}^2 + S_{22}^2 = 1/2 \quad \text{— (2)} \end{array}\right\} S_{11} = S_{22}$$

→ $R_1C_3 \Rightarrow S_{11}S_{13}^* + S_{12}S_{13}^* = 0 \Rightarrow \boxed{S_{11} = -S_{12}}$

→ From eqn (1)

$\Rightarrow s_{11}^2 + s_{11}^2 = 1/2$

$\Rightarrow s_{11}^2 = 1/4$

$\Rightarrow \boxed{S_{11} = 1/2}$

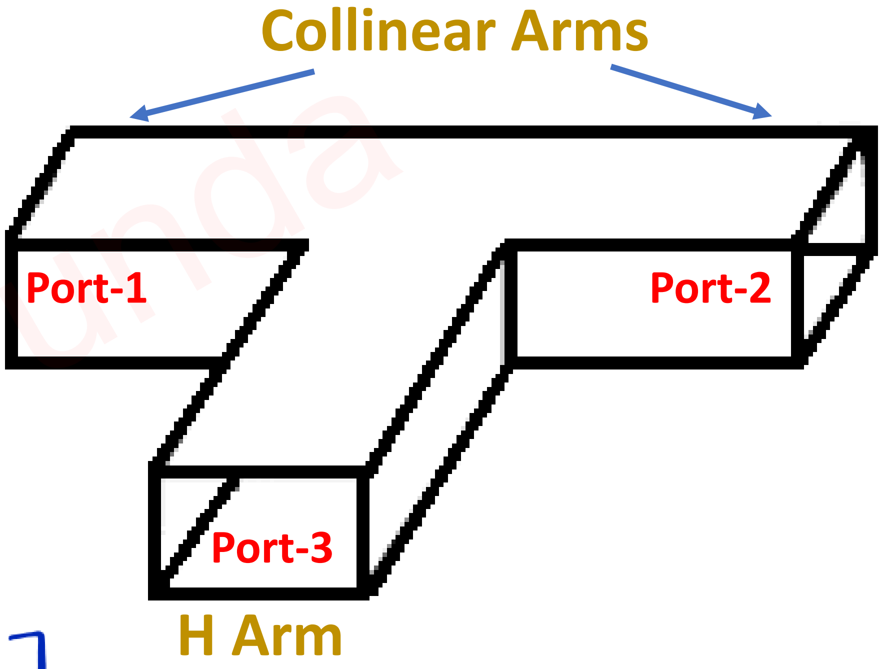
→ So, Scattering matrix of H-plane Tee is

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ -1/2 & 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$$

Applications of H – Plane Tee Junction

- Power Divider and Power Combiner
- Microwave Signal Routing
- Microwave Mixers
- Antenna Feed Network
- Measurement and Testing of Microwave Circuits

Case Study of H – Plane Tee



⤳ S Matrix of H-plane Tee junction is given by

$$[S] = \begin{vmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ -1/2 & 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{vmatrix}$$

⤳ We know that

$$S = b/a \Rightarrow [b] = [S][a]$$

$$\Rightarrow \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ -1/2 & 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

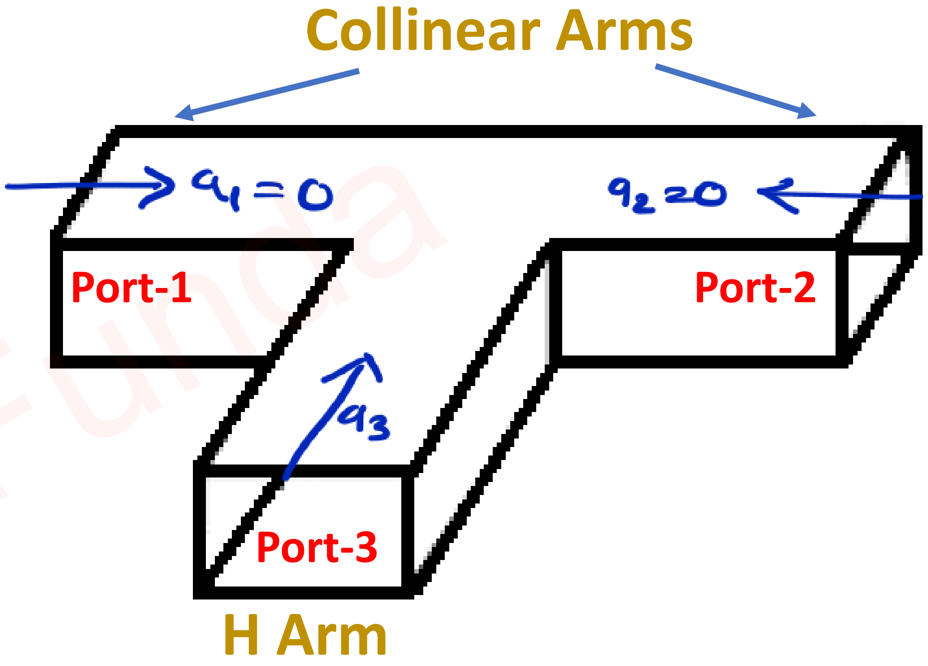
⤳ Output at P-1, P-2 & P-3

$$b_1 = \frac{1}{2}a_1 - \frac{1}{2}a_2 + \frac{1}{\sqrt{2}}a_3$$

$$b_2 = -\frac{1}{2}a_1 + \frac{1}{2}a_2 + \frac{1}{\sqrt{2}}a_3$$

$$b_3 = \frac{1}{\sqrt{2}}a_1 + \frac{1}{\sqrt{2}}a_2$$

Case I: Input is given at port 3 and no inputs are given at Port 1 and Port 2



- $a_1 = a_2 = 0$

  $a_3 \neq 0$

$$b_1 = \frac{1}{2}a_1 - \frac{1}{2}a_2 + \frac{1}{\sqrt{2}}a_3 = \frac{1}{\sqrt{2}}a_3$$

$$b_2 = -\frac{1}{2}a_1 + \frac{1}{2}a_2 + \frac{1}{\sqrt{2}}a_3 = \frac{1}{\sqrt{2}}a_3$$

$$b_3 = \frac{1}{\sqrt{2}}a_1 + \frac{1}{\sqrt{2}}a_2 = 0$$

- Hence, equal power is coming out from port 1 and port 2. Hence, H plane Tee junction is working as 3-dB splitter.

Case II: Inputs are given at Port 1 and Port 2 and no input is given at Port 3

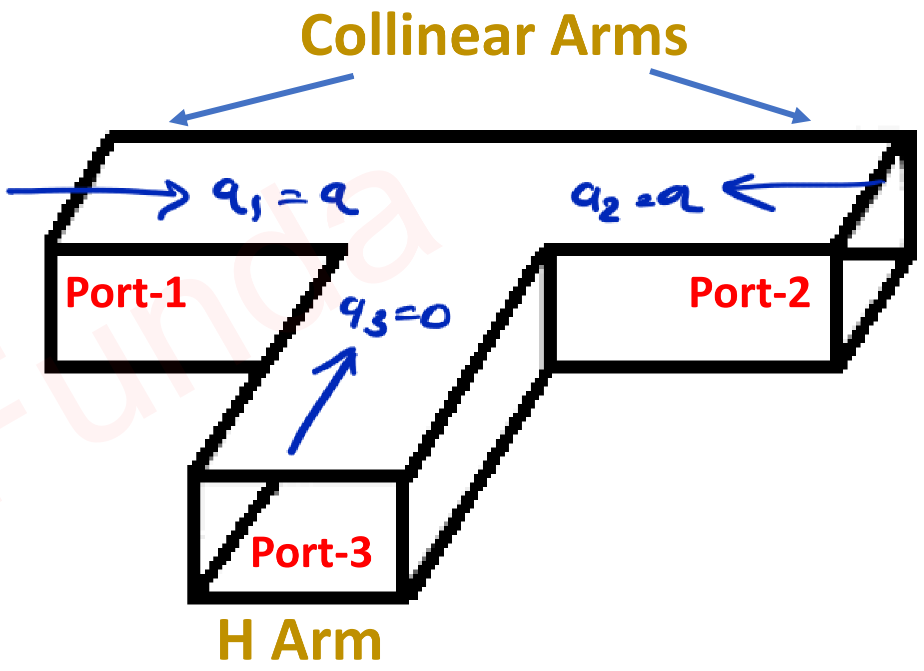
→ $a_1 = a_2 = a$

$a_3 = 0$

$$b_1 = \frac{1}{2}a_1 - \frac{1}{2}a_2 + \frac{1}{\sqrt{2}}a_3 = 0$$

$$b_2 = -\frac{1}{2}a_1 + \frac{1}{2}a_2 + \frac{1}{\sqrt{2}}a_3 = 0$$

$$b_3 = \frac{1}{\sqrt{2}}a_1 + \frac{1}{\sqrt{2}}a_2 = 2\left(\frac{1}{\sqrt{2}}a\right)$$



→ Output at port 3 is addition of power at P-1 and P-2. These power is added in phase.

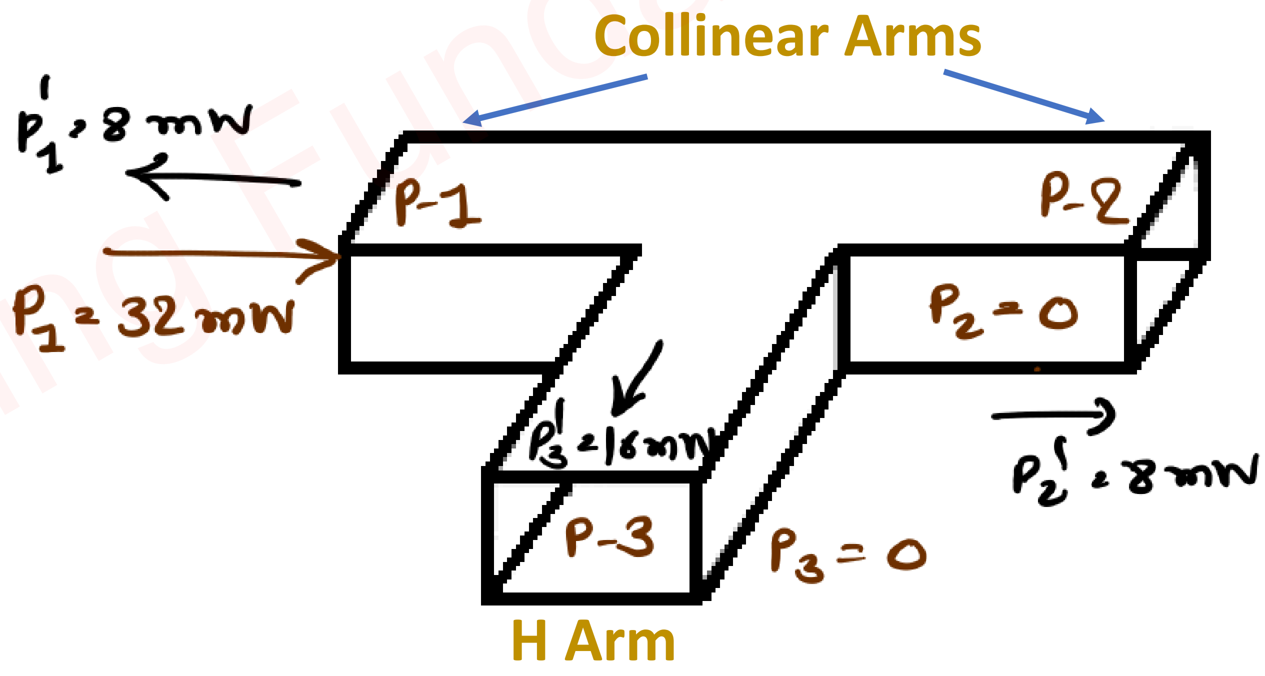
Example of H – Plane Tee

Question 1 – A signal of power 32mW is fed into one of the collinear ports of a lossless H-Plane Tee. Determine the power in the remaining ports when other ports are terminated employing matched loads.

$\Rightarrow P_1 = 32\,mW \qquad \Rightarrow P_1' = ?$

$P_2 = 0 \qquad P_2' = ?$

$P_3 = 0 \qquad P_3' = ?$



$$\Rightarrow [S] = \begin{bmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ -1/2 & 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$$

$$= \frac{[b]}{[a]} = \frac{\text{Normalized O/p Volt.}}{\text{Normalized I/p Volt.}}$$

$$\Rightarrow [b] = [S][a]$$

$$[P_0] = [S^2][P_{in}]$$

$$\Rightarrow \begin{bmatrix} P_1' \\ P_2' \\ P_3' \end{bmatrix} = \begin{bmatrix} (1/2)^2 & (-1/2)^2 & (1/\sqrt{2})^2 \\ (-1/2)^2 & (1/2)^2 & (1/\sqrt{2})^2 \\ (1/\sqrt{2})^2 & (1/\sqrt{2})^2 & 0 \end{bmatrix} \begin{bmatrix} 32\,mW \\ 0 \\ 0 \end{bmatrix}$$

$$P_1' = (1/2)^2 \times 32 = 8\,mW$$

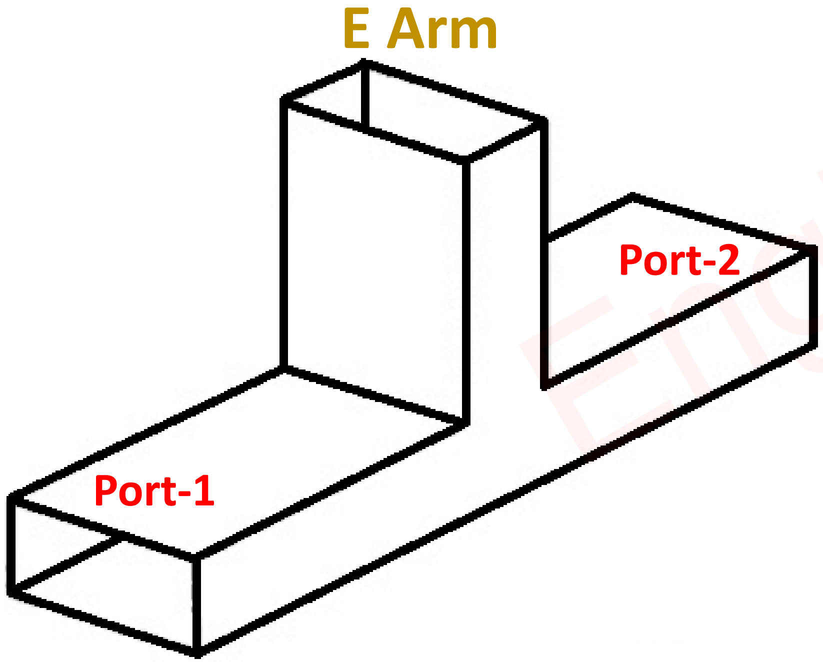
$$P_2' = (-1/2)^2 \times 32 = 8\,mW$$

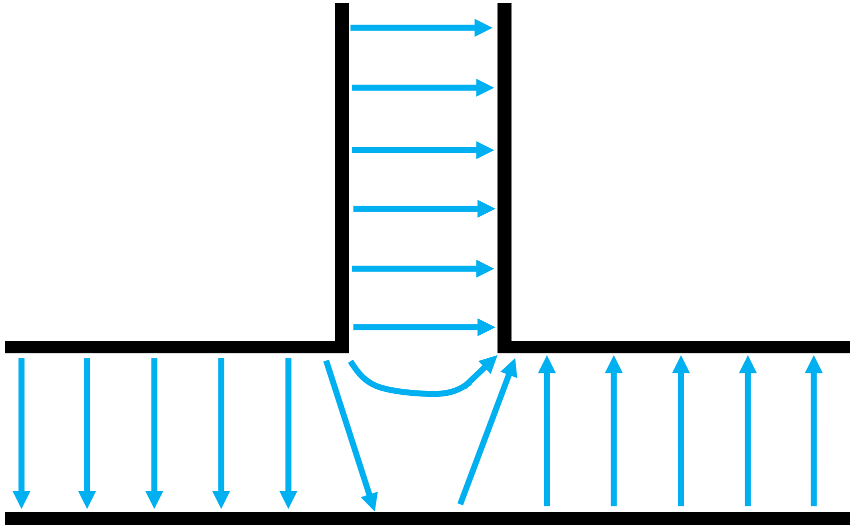
$$P_3' = (1/\sqrt{2})^2 \times 32 = 16\,mW$$

E – Plane Tee Junction

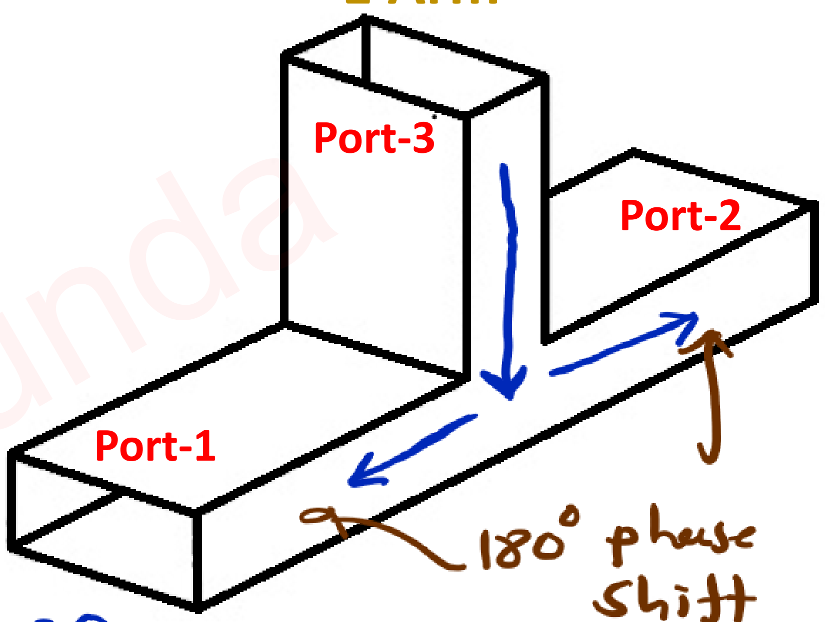
Basics of E – Plane Tee Junction

- E – Plane Tee Junction is a Three port device.
- E – Plane Tee Junction has one E Arm (Port) and Two Ports (Port-1 and Port-2).
- Port-1 and Port-2 are Symmetric with respect to E Arm.
- E – Arm is perfectly matched arm.
- So, If Input is given to E-Arm, then Output at Port-1 and Port-2 is equal and Out of Phase.





Scattering Parameters of E – Plane Tee Junction



- E-plane Tee junction is 3-port device.

  So, S-Matrix is given by :

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

$$S_{ij} = \frac{b}{a}$$

ith o/p Port, jth i/p Port

- E-Arm (Port 3) is perfectly Matched Arm, $S_{33} = 0$
- If Input is given to E-Arm [Port-3], Output at port-1 & port-2 is equal in magnitude and Out of phase (180° phase shift).

$$S_{13} = -S_{23}$$

- As per property of Symmetry, $S_{ij} = S_{ji}$

$$S_{12} = S_{21} \quad , \quad S_{13} = S_{31} = -S_{23} = -S_{32}$$

→ So, S-Matrix will be.

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix}$$

→ As per Identity property, $[S][S]^* = [I]$

$$\Rightarrow \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \times \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & -S_{13}^* \\ S_{13}^* & -S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ $R_3C_3 \Rightarrow S_{13}^2 + S_{13}^2 = 1 \Rightarrow S_{13} = 1/\sqrt{2}$

$$\left.\begin{array}{l} \rightarrow R_1C_1 \Rightarrow S_{11}^2 + S_{12}^2 + S_{13}^2 = 1 \Rightarrow S_{11}^2 + S_{12}^2 = 1/2 \quad \text{--- (1)} \\ \rightarrow R_2C_2 \Rightarrow S_{12}^2 + S_{22}^2 + S_{13}^2 = 1 \Rightarrow S_{12}^2 + S_{22}^2 = 1/2 \quad \text{--- (2)} \end{array}\right\} S_{11} = S_{22}$$

→ $R_1C_3 \Rightarrow S_{11}S_{13}^{*} - S_{12}S_{13}^{*} = 0$

$$\Rightarrow \boxed{S_{11} = S_{12}} \quad \text{— (A)}$$

→ Put (A) into eq^n (1)

$$\Rightarrow S_{11}^2 + S_{11}^2 = 1/2$$

$$\Rightarrow S_{11}^2 = 1/4$$

$$\Rightarrow S_{11} = 1/2$$

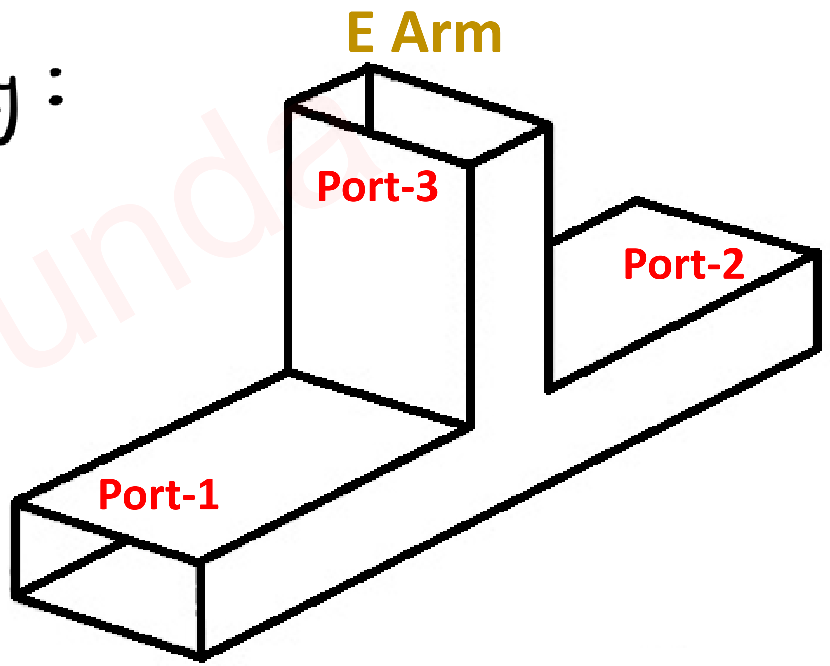
→ S-Matrix of E-plane Tee is given by.

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 1/\sqrt{2} \\ 1/2 & 1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix}$$

Applications of E – Plane Tee Junction

- Power Divider and Power Combiner
- Phase Shifter
- Microwave Mixers
- Impedance Matching Network

Case Study of E – Plane Tee



~ S Matrix of E plane Tee junction is given by :

$$\Rightarrow [S] = \begin{vmatrix} 1/2 & 1/2 & 1/\sqrt{2} \\ 1/2 & 1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{vmatrix} = \frac{[b]}{[a]}$$

$$\Rightarrow [b] = [S][a]$$

$$\Rightarrow \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 1/\sqrt{2} \\ 1/2 & 1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

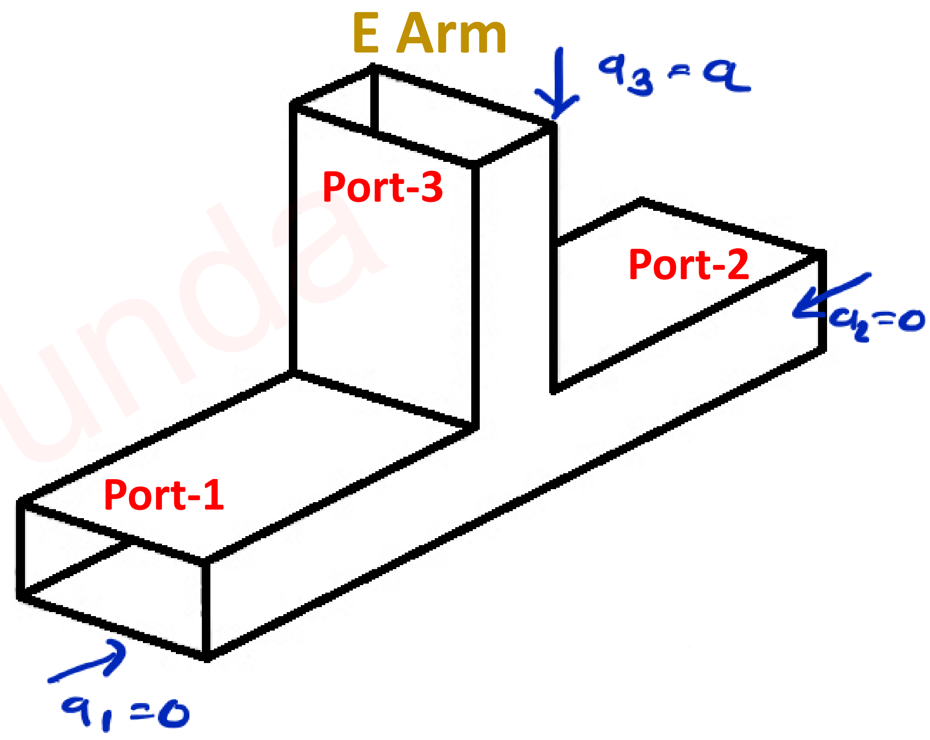
~ So, Output P-1, P-2 and P-3

~ $b_1 = \frac{1}{2} a_1 + \frac{1}{2} a_2 + \frac{1}{\sqrt{2}} a_3$

$b_2 = \frac{1}{2} a_1 + \frac{1}{2} a_2 - \frac{1}{\sqrt{2}} a_3$

$b_3 = \frac{1}{\sqrt{2}} a_1 - \frac{1}{\sqrt{2}} a_2$

Case I: Input is given at port 3 and no inputs are given at Port 1 and Port 2



- $a_1 = a_2 = 0$

  $a_3 \neq 0$

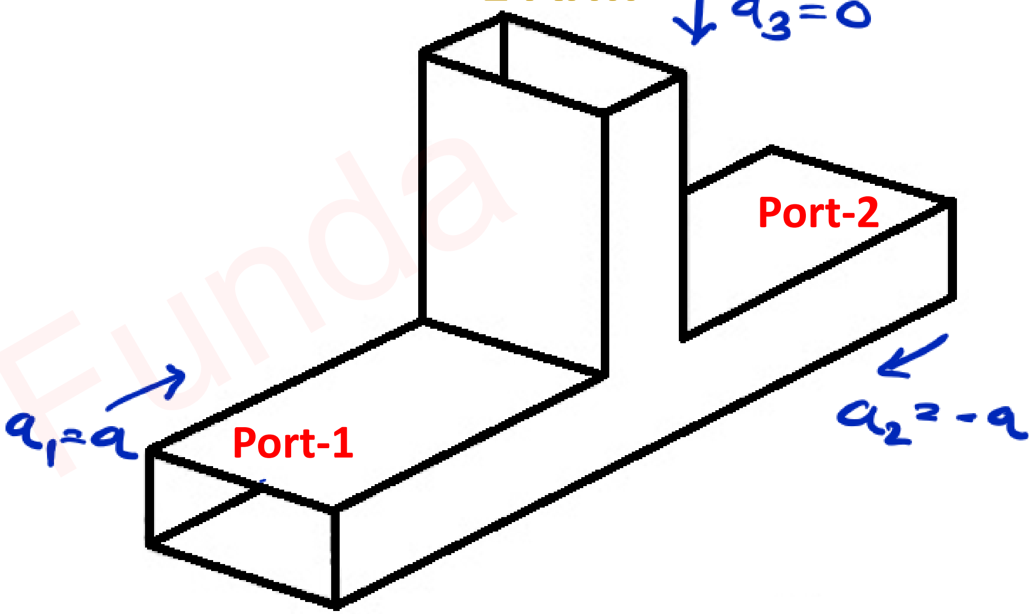
- $b_1 = \frac{1}{2}a_1 + \frac{1}{2}a_2 + \frac{1}{\sqrt{2}}a_3 = \frac{1}{\sqrt{2}}a_3$

  $b_2 = \frac{1}{2}a_1 + \frac{1}{2}a_2 - \frac{1}{\sqrt{2}}a_3 = -\frac{1}{\sqrt{2}}a_3$

  $b_3 = \frac{1}{\sqrt{2}}a_1 - \frac{1}{\sqrt{2}}a_2 = 0$

- When Input is applied at E-Arm, Power at port-1 and Port-2 is equal and Out of phase (180° phase shift).

Case II: Inputs are given at Port 1 and Port 2 (180^0 Out of phase) and No input is given at Port 3



- $a_3 = 0$

  $a_1 = -a_2 = a$

- $b_1 = \frac{1}{2}a_1 + \frac{1}{2}a_2 + \frac{1}{\sqrt{2}}a_3 = 0$

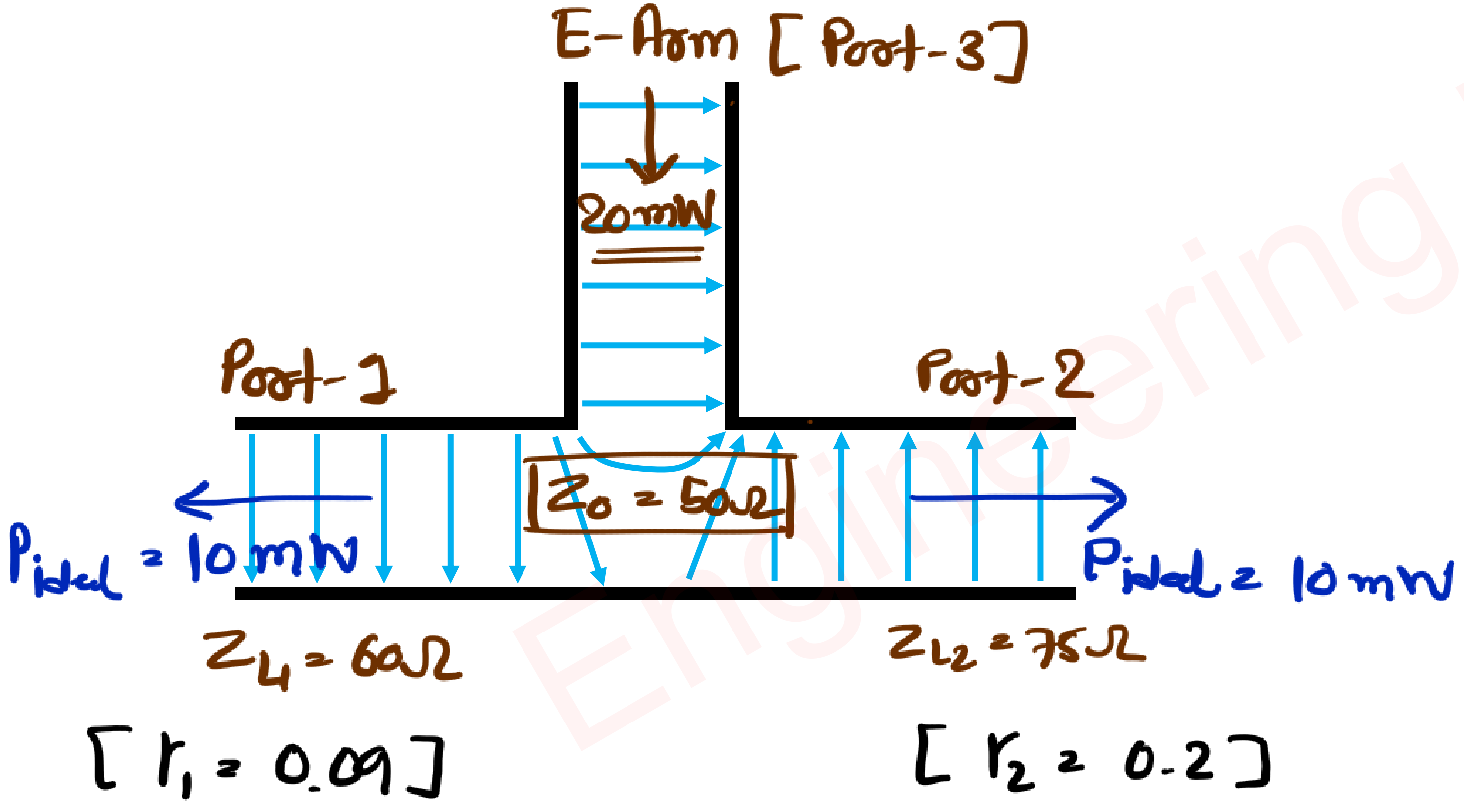
  $b_2 = \frac{1}{2}a_1 + \frac{1}{2}a_2 - \frac{1}{\sqrt{2}}a_3 = 0$

  $b_3 = \frac{1}{\sqrt{2}}a_1 - \frac{1}{\sqrt{2}}a_2 = 2\left(\frac{1}{\sqrt{2}}a\right)$

- If Input at Port-1 and Port-2 is Out of phase, then these I/P's are added at Port-3.

Example of E – Plane Tee

Question 1 – In an E-Plane Tee Junction, 20mW power is applied to Port-3, which perfectly matches the junction. Calculate the power delivered to the load of 60Ω and 75Ω, connected at Port-1 and Port-2, Respectively. $[Z_0 = 50\Omega]$



$$\Gamma_1 = \frac{Z_{L1} - Z_0}{Z_{L1} + Z_0} = \frac{60 - 50}{60 + 50} = 0.09$$

$$\Gamma_2 = \frac{Z_{L2} - Z_0}{Z_{L2} + Z_0} = \frac{75 - 50}{75 + 50} = 0.2$$

$$P_{01} = P_{ideal}\,(1 - |\Gamma_1^2|)$$

$$= 10\,mW\,(1 - 0.09^2)$$

$$= 9.92\,mW$$

$$P_{02} = P_{ideal}\,(1 - |\Gamma_2^2|)$$

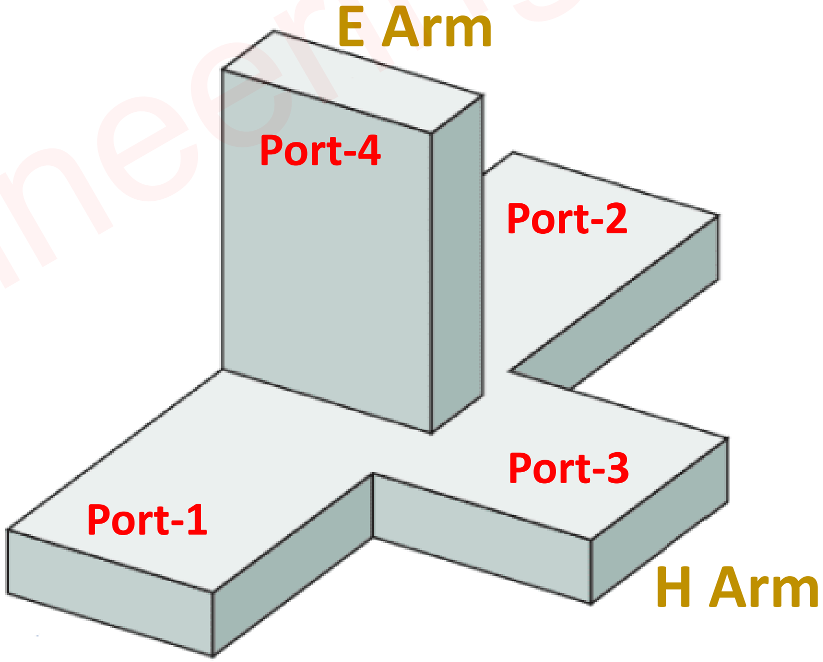
$$= 10\,mW\,(1 - 0.2^2)$$

$$= 9.6\,mW$$

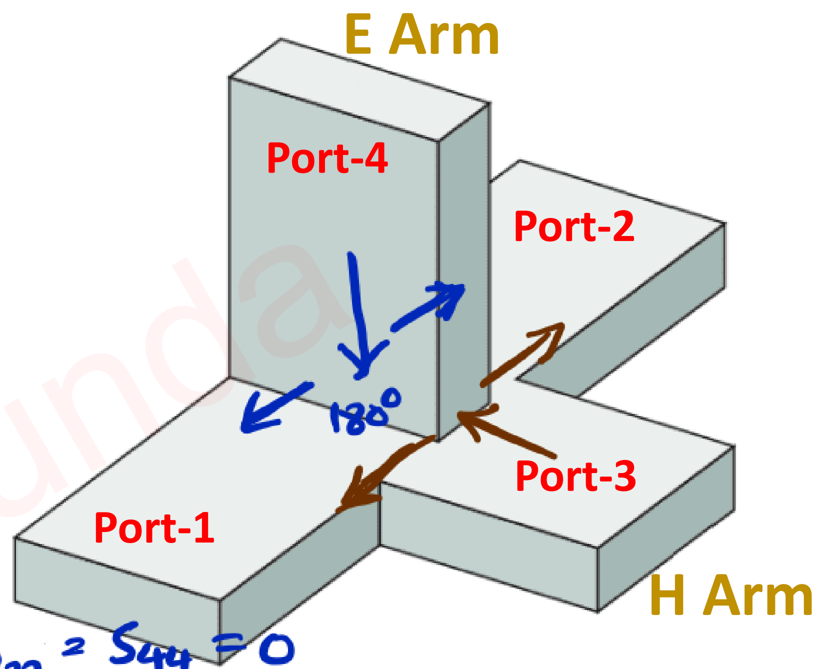
Magic Tee Junction

Basics of Magic Tee Junction

- Magic Tee Junction is a Four-Port device.
- Port-1 and Port-2 are Symmetric with respect to E-Arm and H-Arm.
- E – Arm and H-Arm are perfectly matched Arms and Isolated from each other.
- So, If Input is given to E-Arm, then Output at Port-1 and Port-2 is equal and Out of Phase.
- So, If Input is given to H-Arm, then Output at Port-1 and Port-2 is equal and in Phase.



Scattering Parameters of Magic Tee Junction



- Magic Tee junction is 4-Port device. So S-Matrix Size is (4 × 4).

$$[S] = \begin{vmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{vmatrix}$$

$$S_{ij} = \frac{b}{a}$$

i → O/p port, j → I/p port

- E-Arm and H-Arm are perfectly Matched Arms, $S_{33} = S_{44} = 0$
- E-Arm and H-Arm are isolated with each other, $S_{34} = S_{43} = 0$
- E-Arm property, $S_{14} = -S_{24}$
- H-Arm property, $S_{13} = S_{23}$
- Symmetry property, $S_{ij} = S_{ji}$

  - $S_{12} = S_{21}$, $S_{13} = S_{31} = S_{23} = S_{32}$, $S_{14} = S_{41} = -S_{24} = -S_{42}$

$\rightsquigarrow$ So, S-Matrix is given by

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

$\rightsquigarrow$ By, Applying Identity property, $[S][S]^* = [I]$

$$\Rightarrow \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* & S_{14}^* \\ S_{12}^* & S_{22}^* & S_{13}^* & -S_{14}^* \\ S_{13}^* & S_{13}^* & 0 & 0 \\ S_{14}^* & -S_{14}^* & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\rightsquigarrow$ $R_4C_4 \Rightarrow S_{14}^2 + S_{14}^2 = 1 \Rightarrow S_{14} = 1/\sqrt{2}$

$\rightsquigarrow$ $R_3C_3 \Rightarrow S_{13}^2 + S_{13}^2 = 1 \Rightarrow S_{13} = 1/\sqrt{2}$

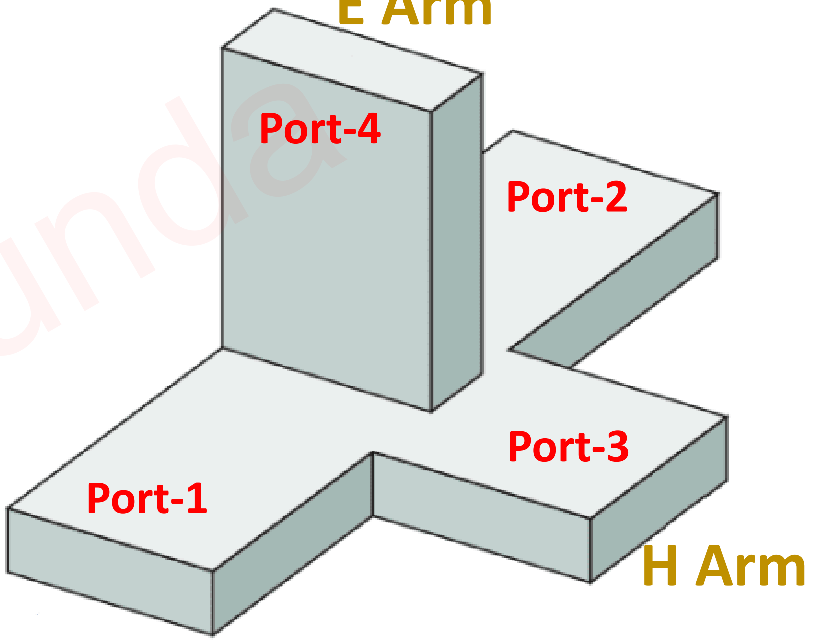
$\rightsquigarrow$ $R_2C_2 \Rightarrow S_{12}^2 + S_{22}^2 + S_{13}^2 + S_{14}^2 = 1 \Rightarrow S_{12}^2 + S_{22}^2 = 0 \Big\} \; S_{12} = 0, \; S_{22} = 0$

$\rightsquigarrow$ $R_1C_1 \Rightarrow S_{11}^2 + S_{12}^2 + S_{13}^2 + S_{14}^2 = 1 \Rightarrow S_{11}^2 + S_{12}^2 = 0 \Big\} \; S_{11} = 0, \; S_{12} = 0$

→ So, S-Matrix is given by

$$[S] = \begin{vmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \end{vmatrix}$$

Case Study of Magic Tee



→ S Matrix of Magic Tee given by

$$\Rightarrow [S] = \begin{bmatrix} 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \end{bmatrix} = \frac{[b]}{[a]}$$

$$\Rightarrow [b] = [S][a]$$

$$\Rightarrow \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

→ Output voltage eq^{n}s are given by:

→ $b_1 = \frac{1}{\sqrt{2}}(a_3 + a_4)$

→ $b_2 = \frac{1}{\sqrt{2}}(a_3 - a_4)$

→ $b_3 = \frac{1}{\sqrt{2}}(a_1 + a_2)$

→ $b_4 = \frac{1}{\sqrt{2}}(a_1 - a_2)$

Case I: Input is given at Port 3 and no inputs are given at Port 1, Port 2 & Port 4

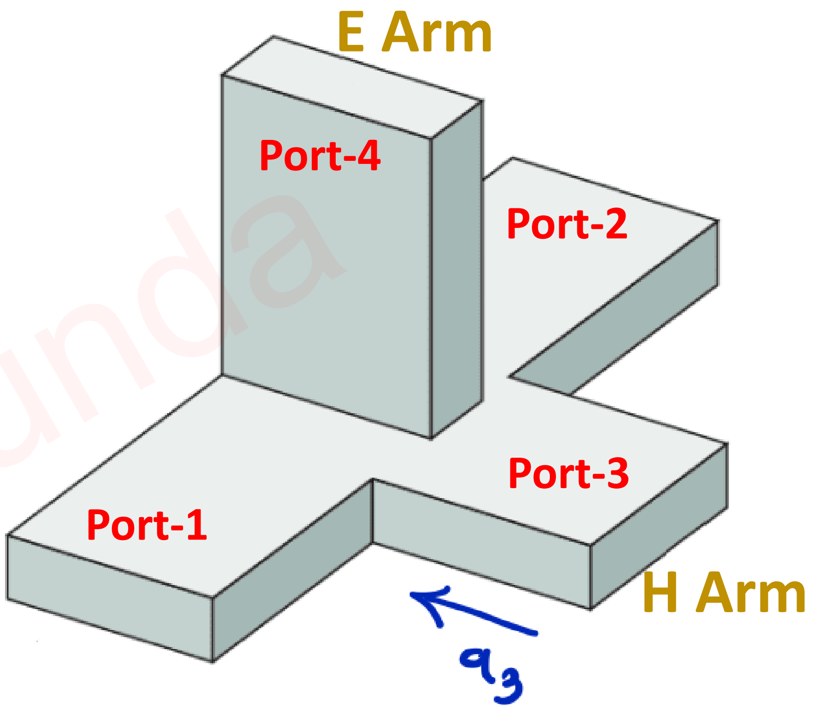
$\rightsquigarrow a_3 \neq 0, \quad a_1 = a_2 = a_4 = 0$

$\rightsquigarrow b_1 = \frac{1}{\sqrt{2}}(a_3 + a_4) = \frac{1}{\sqrt{2}} a_3$

$\rightsquigarrow b_2 = \frac{1}{\sqrt{2}}(a_3 - a_4) = \frac{1}{\sqrt{2}} a_3$

$\rightsquigarrow b_3 = \frac{1}{\sqrt{2}}(a_1 + a_2) = 0$

$\rightsquigarrow b_4 = \frac{1}{\sqrt{2}}(a_1 - a_2) = 0$



$\rightsquigarrow$ So, If Input is given to H-arm, O/p at p-1 & p-2 is equal and In phase (0° phase shift).

Case II: Input is given at Port 4 and no inputs are given at Port 1, Port 2 & Port 3

→ $a_4 \neq 0$, $a_1 = a_2 = a_3 = 0$

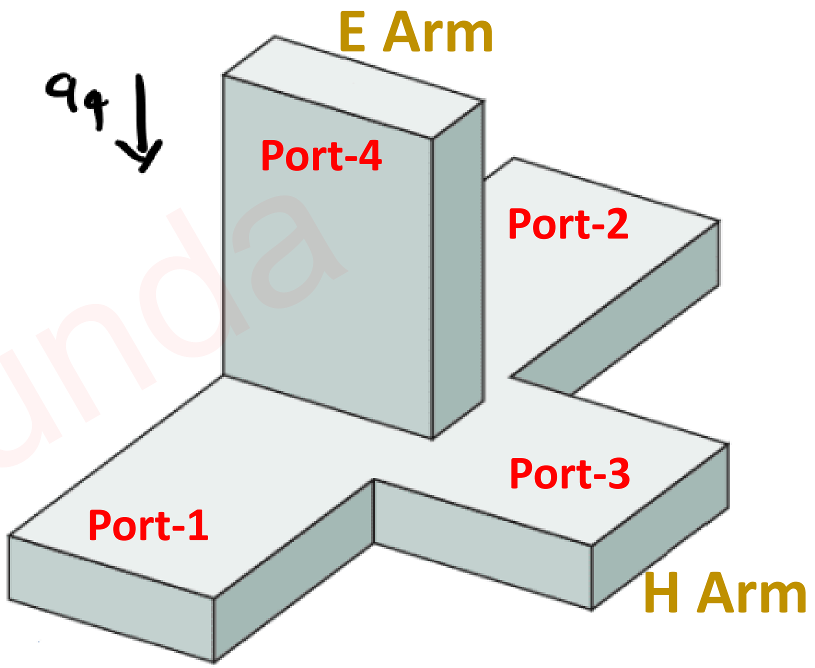
→ $b_1 = \frac{1}{\sqrt{2}}(a_3 + a_4) = \frac{1}{\sqrt{2}} a_4$

→ $b_2 = \frac{1}{\sqrt{2}}(a_3 - a_4) = -\frac{1}{\sqrt{2}} a_4$

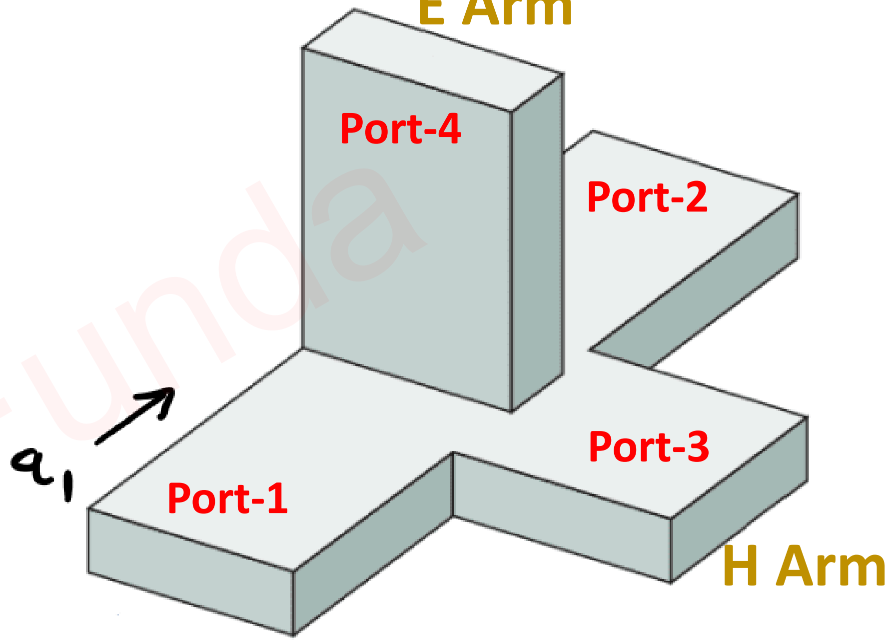
→ $b_3 = \frac{1}{\sqrt{2}}(a_1 + a_2) = 0$

→ $b_4 = \frac{1}{\sqrt{2}}(a_1 - a_2) = 0$

→ So, If Input is given to E-arm, Output at P-1 & P-2 is equal and Out of phase (180° phase shift).



Case III: Input is given at Port 1 and no inputs are given at Port 2, Port 3 & Port 4

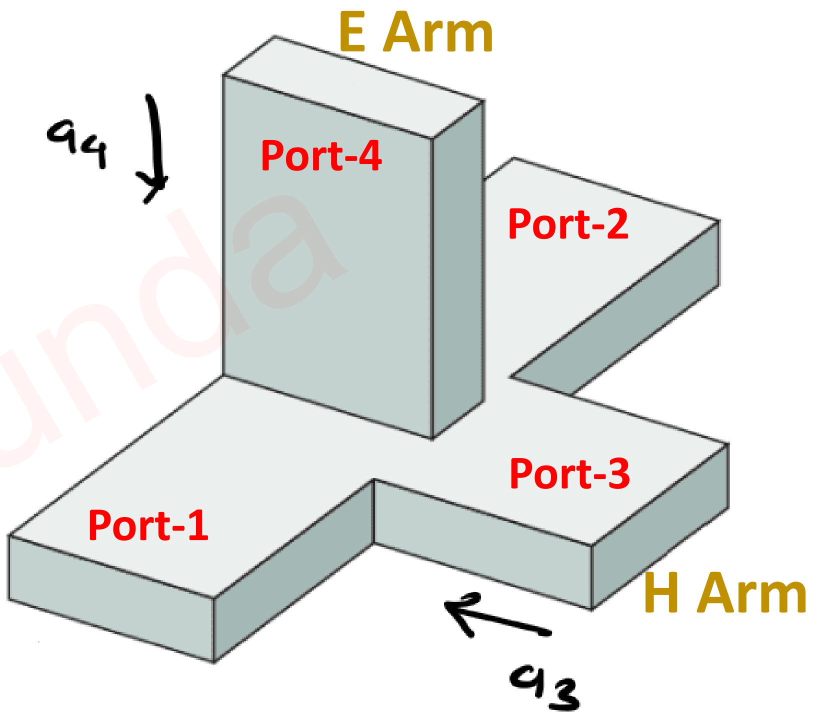


- $a_1 \neq 0$, $a_2 = a_3 = a_4 = 0$

  - $b_1 = \frac{1}{\sqrt{2}}(a_3 + a_4) = 0$
  - $b_2 = \frac{1}{\sqrt{2}}(a_3 - a_4) = 0$
  - $b_3 = \frac{1}{\sqrt{2}}(a_1 + a_2) = \frac{1}{\sqrt{2}} a_1$
  - $b_4 = \frac{1}{\sqrt{2}}(a_1 - a_2) = \frac{1}{\sqrt{2}} a_1$

- If Input is given to P-1, It is equally divided Into E-Arm & H-Arm and O/p at Colinear arm (P-2) is zero.

Case IV: Input is given at Port 3 & Port 4 and no inputs are given at Port 1 & Port 2

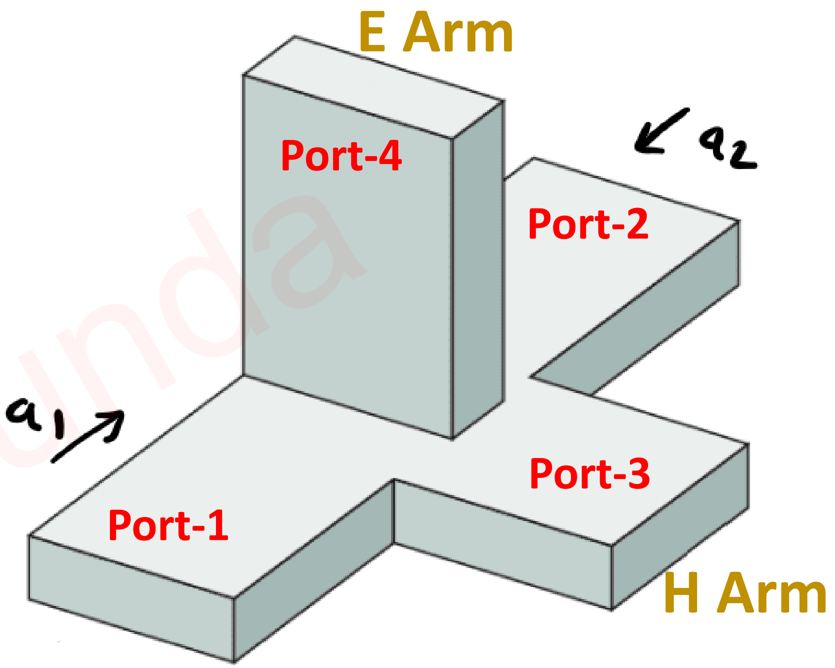


- $a_3 = a_4 = a \quad , \quad a_1 = a_2 = 0$

  - $b_1 = \frac{1}{\sqrt{2}}(a_3 + a_4) = 2\left(\frac{1}{\sqrt{2}}a\right)$
  - $b_2 = \frac{1}{\sqrt{2}}(a_3 - a_4) = 0$
  - $b_3 = \frac{1}{\sqrt{2}}(a_1 + a_2) = 0$
  - $b_4 = \frac{1}{\sqrt{2}}(a_1 - a_2) = 0$

- It is additive property of magic Tee.

Case V: Input is given at Port 1 & Port 2 and no inputs are given at Port 3 & Port 4



→ $a_1 = a_2 = a$, $a_3 = a_4 = 0$

→ $b_1 = \frac{1}{\sqrt{2}}(a_3 + a_4) = 0$

→ $b_2 = \frac{1}{\sqrt{2}}(a_3 - a_4) = 0$

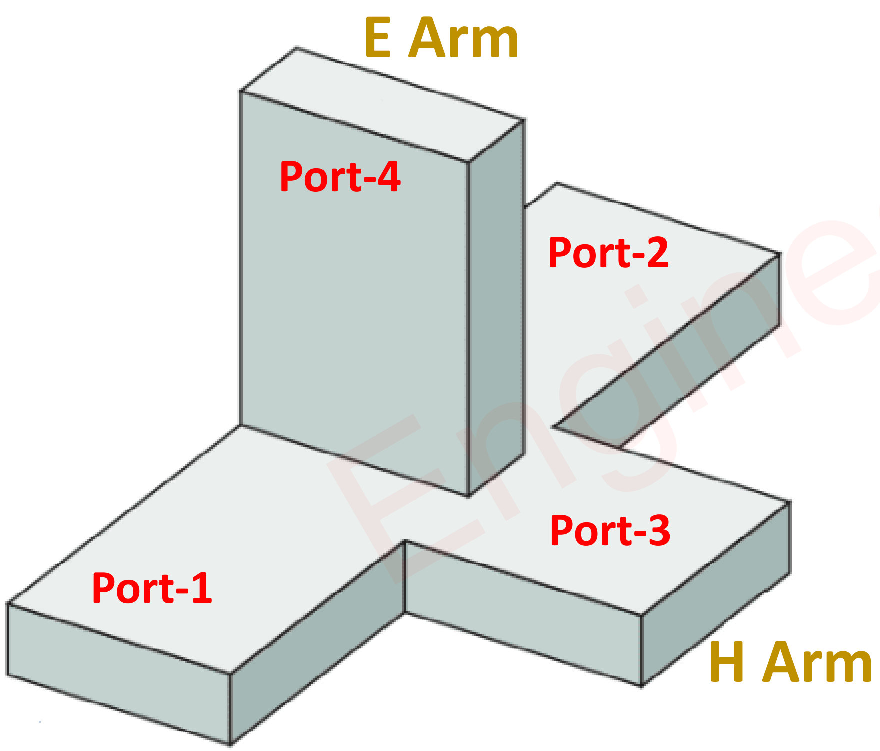
→ $b_3 = \frac{1}{\sqrt{2}}(a_1 + a_2) = 2\left(\frac{1}{\sqrt{2}} a\right)$

→ $b_4 = \frac{1}{\sqrt{2}}(a_1 - a_2) = 0$

→ It is additive property of magic Tee.

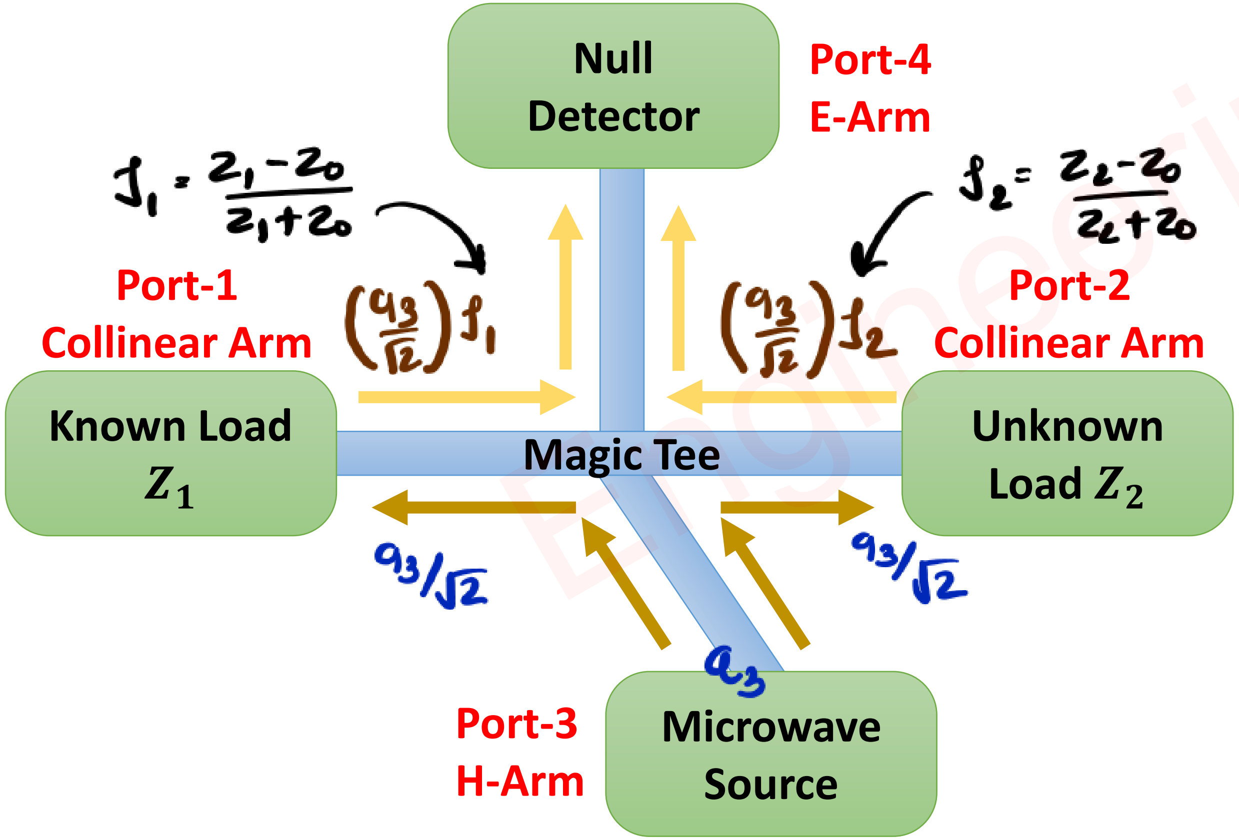
Applications of Magic Tee

- There are three main Applications of Magic Tee Junction
  - Measurement of Impedance
  - Magic Tee as Duplexer
  - Magic Tee as Mixer



❖ Impedance Measurement using Magic Tee Junction

- Microwave Source is connected with H-Arm and Signal from H-Arm will go towards Port-1 {Known Load Z_1} and Port-2 {Unknown Load Z_2}.
- Reflection from Port-1 {Known Load Z_1} and Port-2 {Unknown Load Z_2} will go towards Null Detector {Port-4, E-Arm}.
- By changing value of known load Z_1 {At Port-1}, At $Z_1 = Z_2$, Null Detector shows Null Detection.



- **Output at Port-4 {Null Detector, E-Arm}**

$$\Rightarrow b_4 = \frac{1}{\sqrt{2}}\rho_1 a_3 - \frac{1}{\sqrt{2}}\rho_2 a_3$$

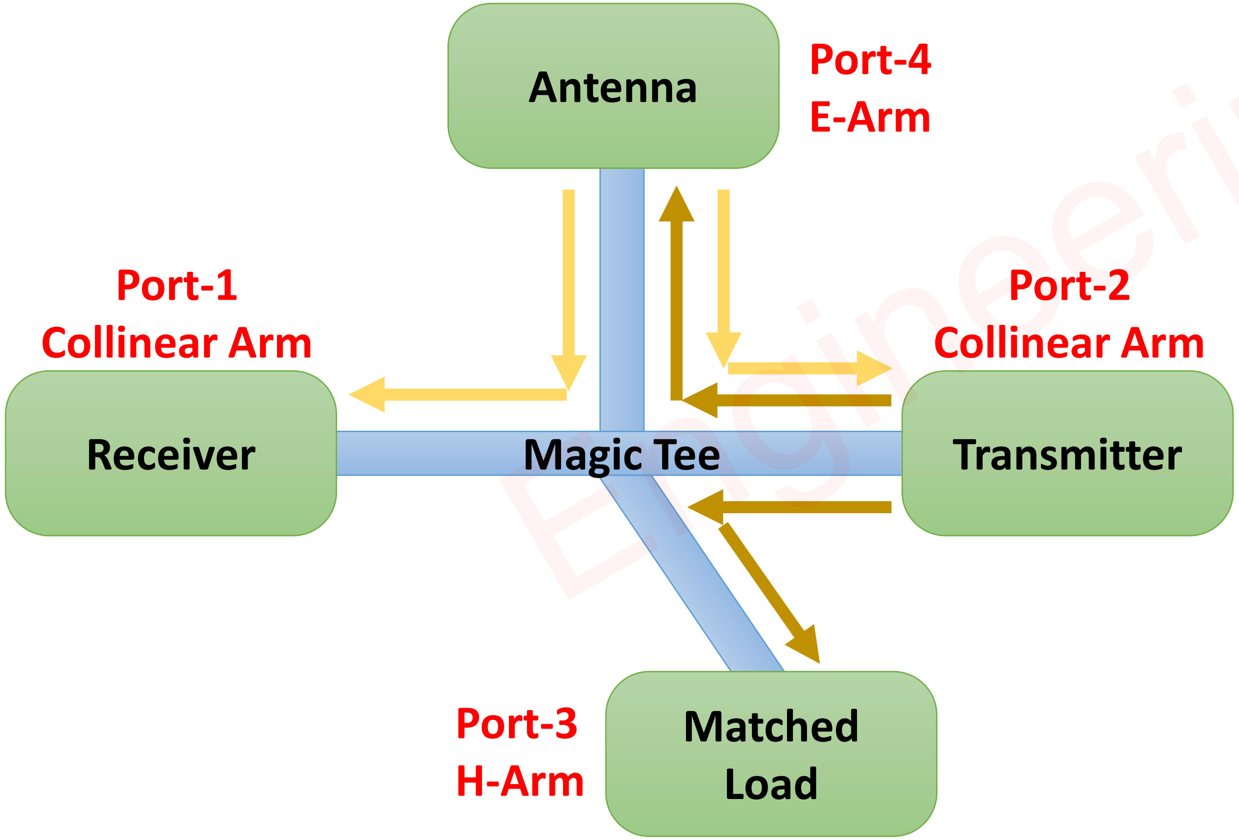
$$\Rightarrow b_4 = \frac{1}{\sqrt{2}} a_3(\rho_1 - \rho_2)$$

$$\Rightarrow b_4 = \frac{1}{\sqrt{2}} a_3\left(\frac{Z_1 - Z_0}{Z_1 + Z_0} - \frac{Z_2 - Z_0}{Z_2 + Z_0}\right)$$

- **At, $Z_1 = Z_2$, $b_4 = 0$, Null Detector shows Zero deflection.**

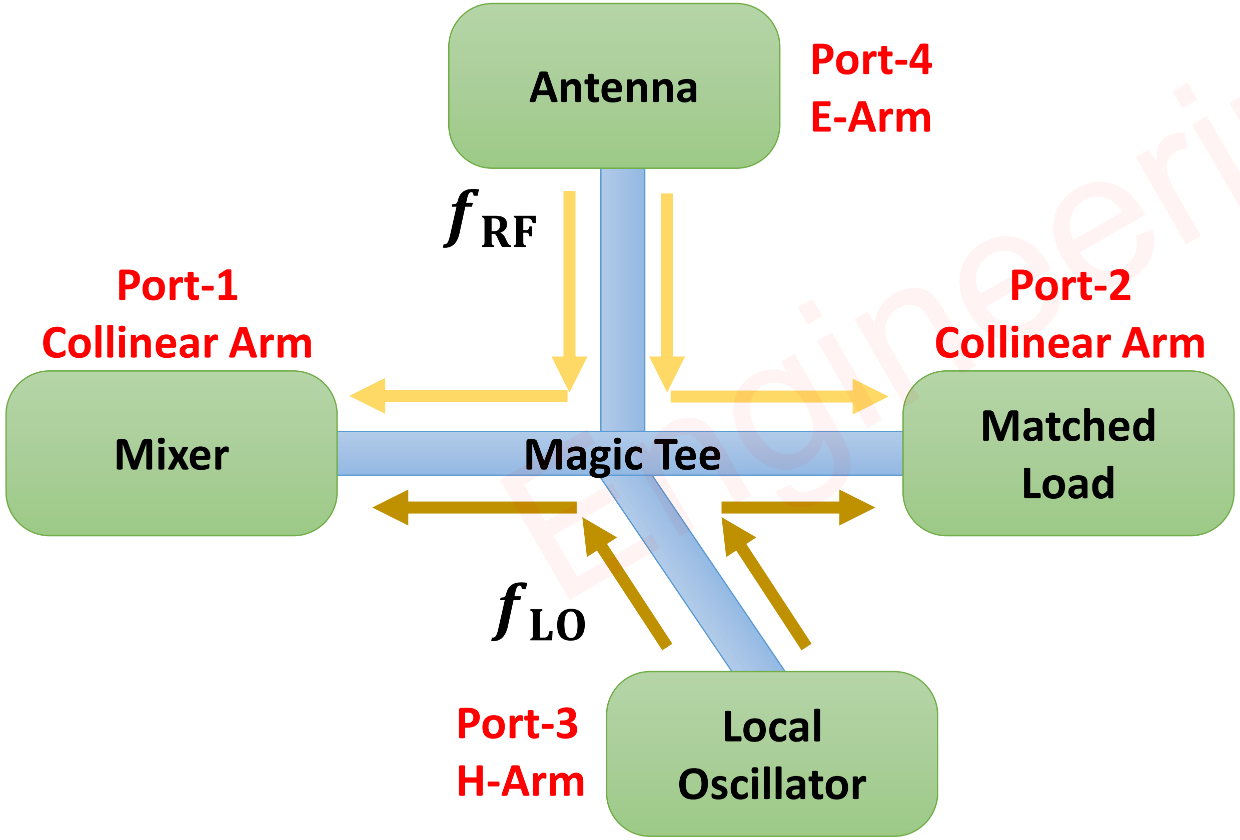
❖ Magic Tee as Duplexer

- Signal of Transmitter {Port-2} will go Towards Antenna {Port-4, E-Arm} and Matched Load {Port-3, H-Arm}.
- Signals Received by Antenna {Port-4, E-Arm} will go Towards Receiver {Port-1} and Transmitter {Port-2}.
- Here Magic Tee is used as a Duplexer to Isolate Low Power {Sensitive} Receiver and Higher Power Transmitter.



❖ Magic Tee as Mixer

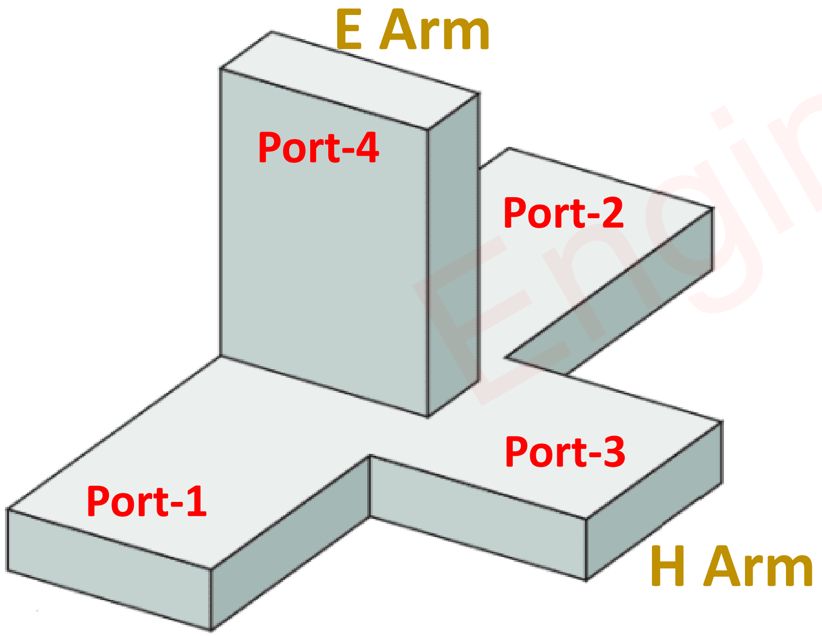
- ❑ Signal of the Local Oscillator will go towards Mixer {Port-1} and Matched load {Port-2}.
- ❑ Signal of the Antenna will go towards Mixer {Port-1} and Matched load {Port-2}.
- ❑ Here Magic Tee is used as a Mixer to Mix Signals of Local Oscillator {Port-3, H-Arm} and Antenna {Port-4, E-Arm}.
- ❑ Mixer will do Addition and Subtraction of frequencies of Local Oscillator {Port-3, H-Arm} and Antenna {Port-4, E-Arm}.



Pros & Cons of Magic Tee

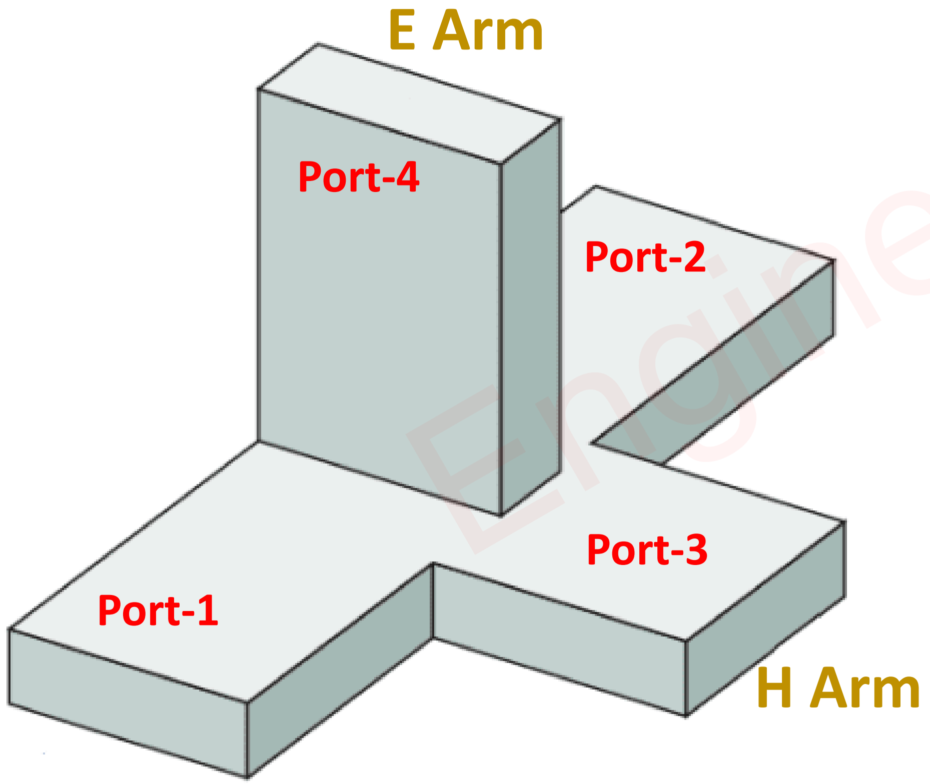
❑ Advantages of Magic Tee Junction

- Output Ports are decoupled to each other {E-Arm and H-Arm/ Port-1 and Port-2}. As a result, power delivered to one of the output ports does not depend on termination at other ports.
- Output ports are perfectly matched. Hence, Power division between them does not depend on terminations connected on any of the ports of Magic Tee unlike E-Plane Tee and H-Plane Tee.
- Magic Tee Junction is Small in Size.
- Magic Tee Junction supports Large Bandwidth.



❑ Disadvantages of Magic Tee Junction

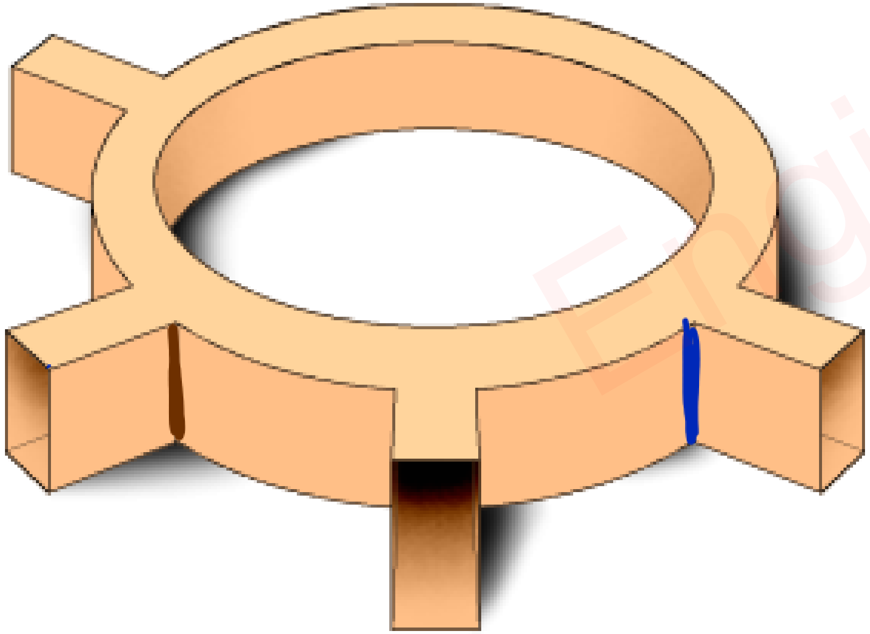
- When Energy propagates through the arm of The Magic Tee, it results in reflections due to impedance mismatch at the junction. This results in power loss.
- Due to reflections of signals, It generates standing patterns inside The Magic Tee waveguide. This limits the power handling capability of the Magic Tee device.
- Lines are very Narrow and close to each other.
- Performance of The Magic Tee significantly depends on Operating Frequency.
- Design Complexity is higher then E-Plane Tee and H-Plane Tee.

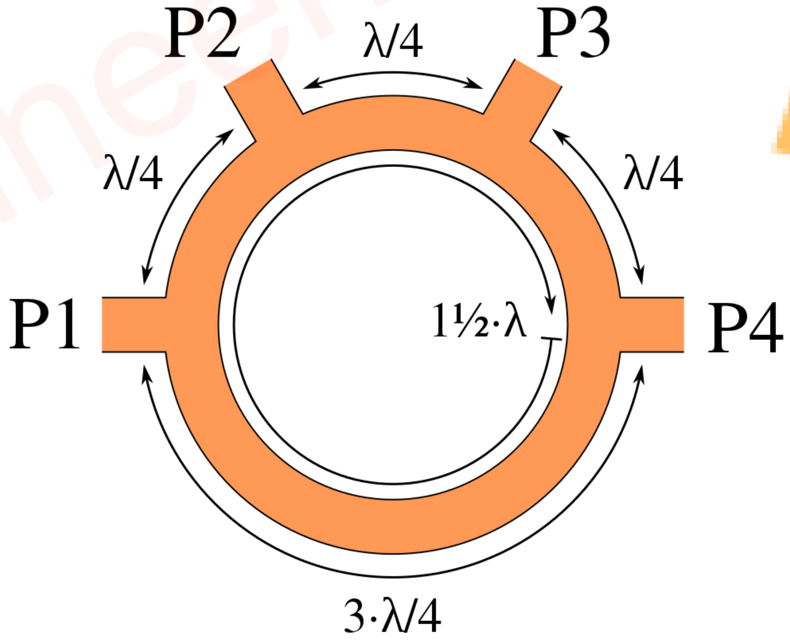


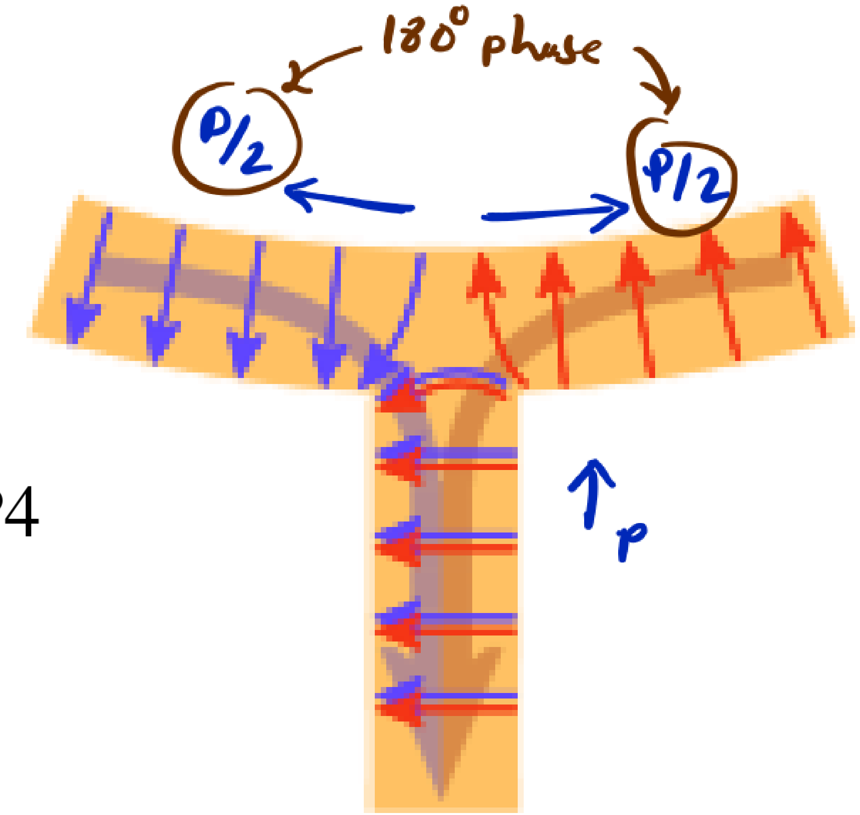
Hybrid Ring Junction

Basics of Hybrid Ring Junction

- Hybrid Ring Junction is also known as Rate-Race Junction.
- Hybrid Ring Junction is constructed by Rectangular waveguide by banding it in a Circle of length 1.5λ.
- Ports 1 & 3 and Ports 2 & 4 are Isolated with Each other.
- All the ports are perfectly matched ports.
- Hybrid Ring Junction consists of Four E-Type T-Junctions.

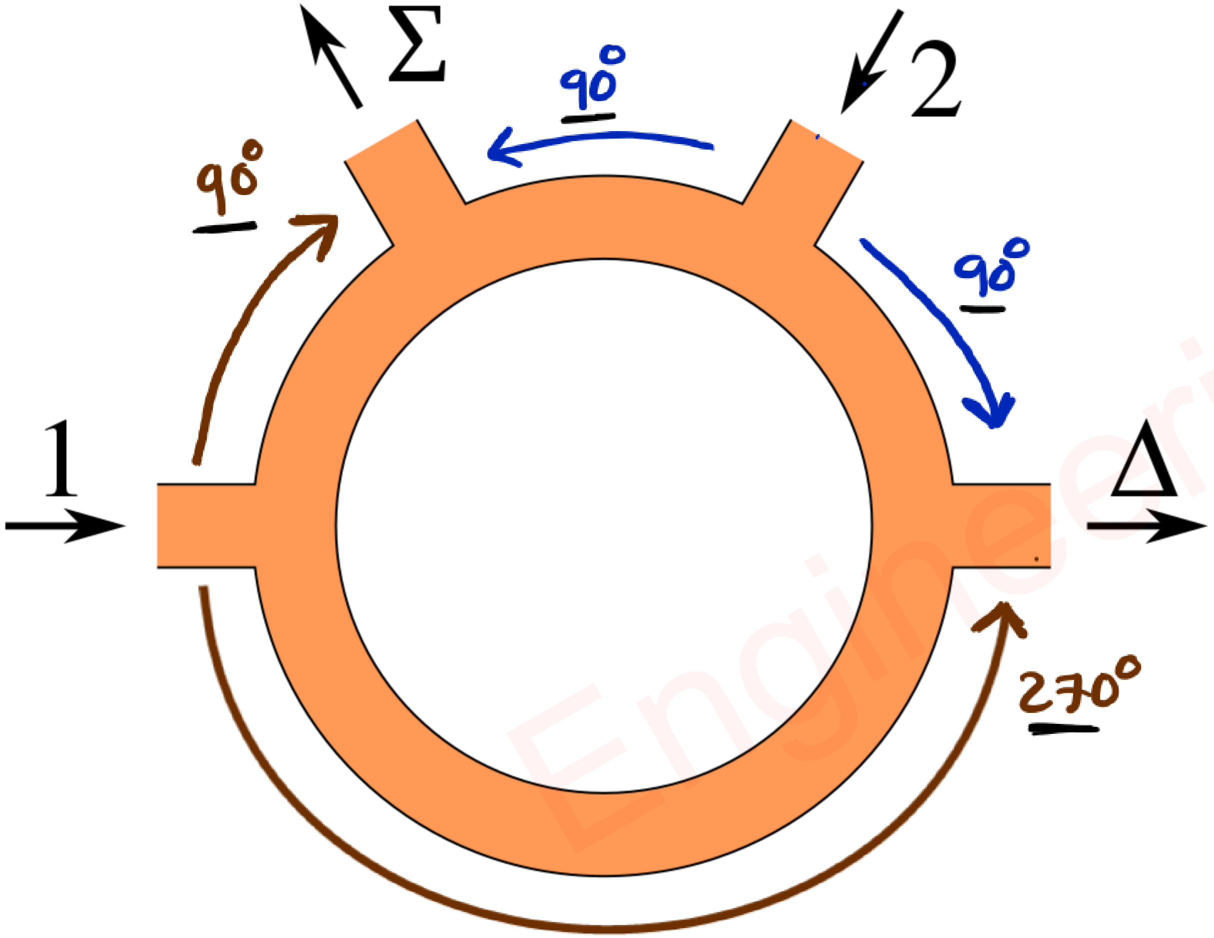


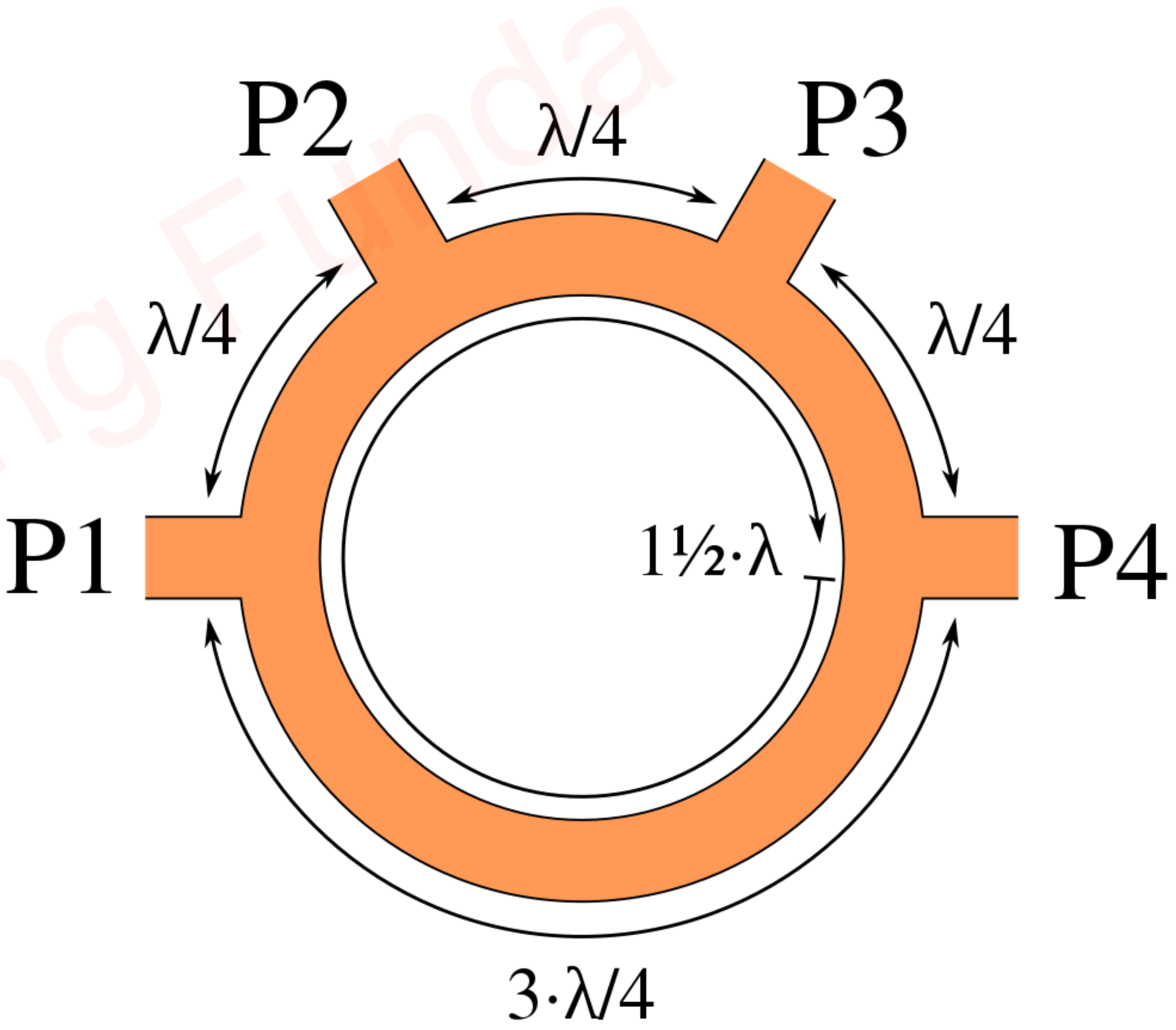




Working of Hybrid Ring Junction

- If Inputs are given at Port-1 and Port-3, Output at Port-2 will be an addition of Inputs and Output at Port-4 will be a Subtraction of Inputs.



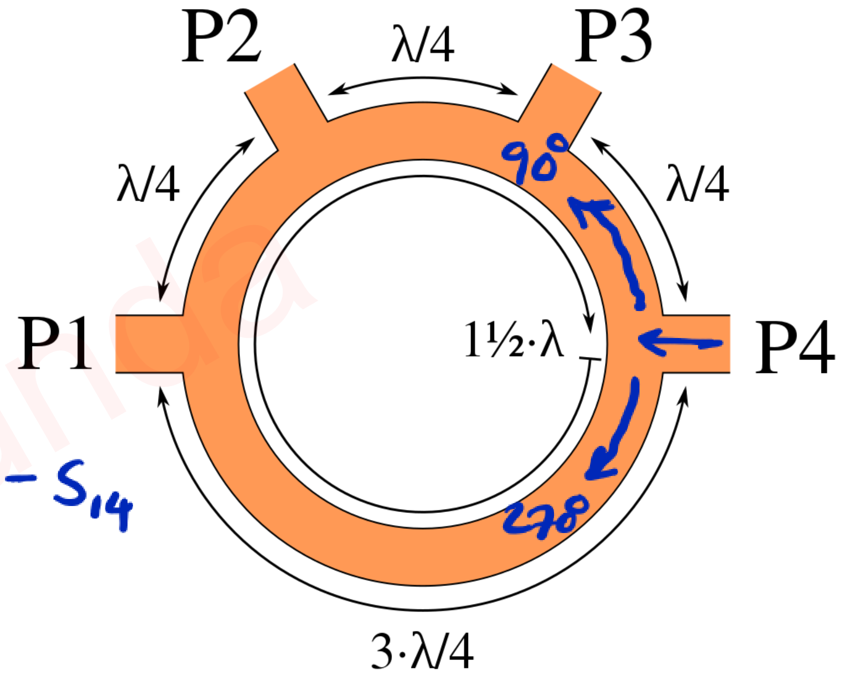


Scattering Parameters of Hybrid Ring Junction

- Hybrid Ring junction is 4-port n/w.
  So, S-Matrix is 4×4 matrix.

$$[S] = \begin{vmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{vmatrix}$$

$S_{34} = -S_{14}$



- All ports are perfectly matched ports. ⇒ $S_{11} = S_{22} = S_{33} = S_{44} = 0$
- Ports 1 & 3 and Ports 2 & 4 are isolated with each other.

$$S_{13} = S_{31} = S_{24} = S_{42} = 0$$

- So, modified Scattering matrix will be.

$$[S] = \begin{vmatrix} 0 & S_{12} & 0 & S_{14} \\ S_{21} & 0 & S_{23} & 0 \\ 0 & S_{32} & 0 & S_{34} \\ S_{41} & 0 & S_{43} & 0 \end{vmatrix} = \begin{vmatrix} 0 & S_{12} & 0 & -S_{34} \\ S_{21} & 0 & S_{23} & 0 \\ 0 & S_{12} & 0 & S_{34} \\ -S_{21} & 0 & S_{23} & 0 \end{vmatrix}$$

→ From Identity property,

$$[S][S]^* = [I]$$

$$\Rightarrow \begin{vmatrix} 0 & S_{12} & 0 & -S_{34} \\ S_{21} & 0 & S_{23} & 0 \\ 0 & S_{12} & 0 & S_{34} \\ -S_{21} & 0 & S_{23} & 0 \end{vmatrix} \begin{vmatrix} 0 & S_{12}^* & 0 & -S_{34}^* \\ S_{21}^* & 0 & S_{23}^* & 0 \\ 0 & S_{12}^* & 0 & S_{34}^* \\ -S_{21}^* & 0 & S_{23}^* & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

→ $C_1 P_1 \Rightarrow S_{21}^2 + S_{21}^2 = 1 \Rightarrow S_{21} = 1/\sqrt{2}$

→ $C_2 P_2 \Rightarrow S_{12}^2 + S_{12}^2 = 1 \Rightarrow S_{12} = 1/\sqrt{2}$

→ $C_3 P_3 \Rightarrow S_{23}^2 + S_{23}^2 = 1 \Rightarrow S_{23} = 1/\sqrt{2}$

→ $C_4 P_4 \Rightarrow S_{34}^2 + S_{34}^2 = 1 \Rightarrow S_{34} = 1/\sqrt{2}$

→ So Scattering Matrix of Hybrid ring junction is given by:

$$[S] = \begin{vmatrix} 0 & S_{12} & 0 & -S_{34} \\ S_{21} & 0 & S_{23} & 0 \\ 0 & S_{12} & 0 & S_{34} \\ -S_{21} & 0 & S_{23} & 0 \end{vmatrix} = \begin{vmatrix} 0 & 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} & 0 \end{vmatrix}$$

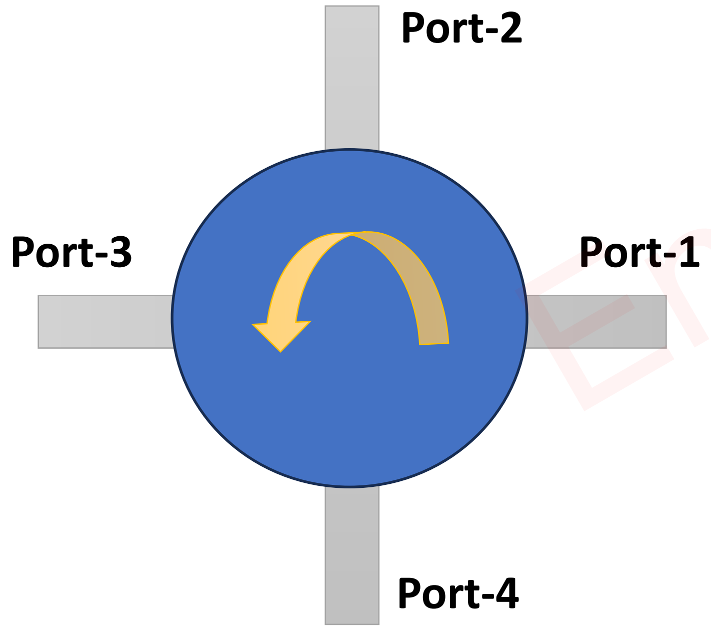
Applications of Hybrid Ring Junction

- Duplexer, Isolator, 3dB Coupler
- Power Divider and Power Combiner
- Mixer
- Impedance Measurement

Circulator Waveguide

Basics of Circulator Waveguide

- A Microwave Circulator is a Multiport Waveguide Junction, In which the wave can flow only from (n)th port to (n+1)th port in one direction.
- Multiport Circulators are available, Most Commonly used circulators are 4-Port Circulators.
- For a given Circulator, If Input is given to Port-1, Output will appear at Port-2 and the remaining ports are isolated.
- If Input is given to Port-2, Output will appear at Port-3 and the remaining ports are isolated.

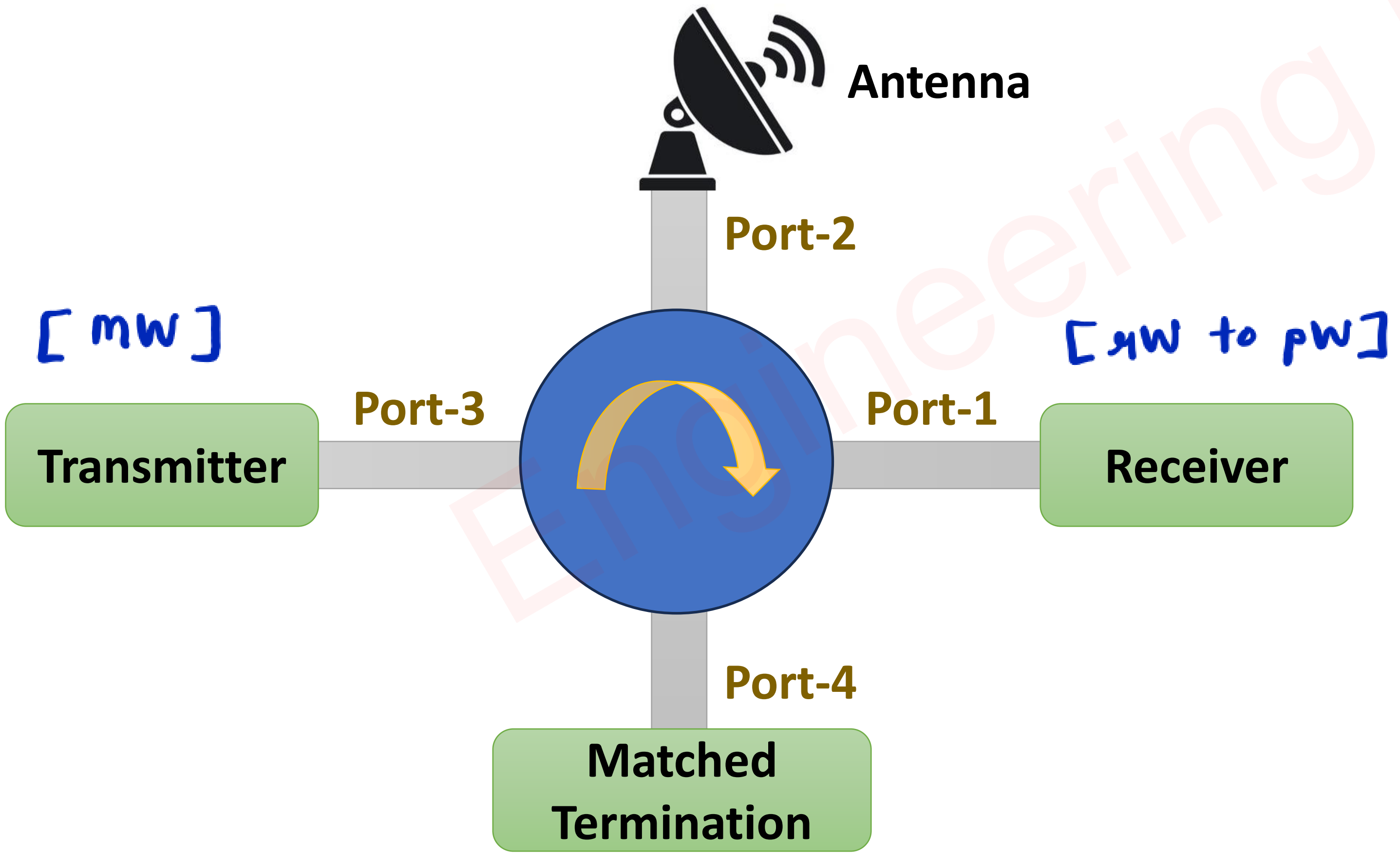


Scattering Matrix of 4-Ports Ideal Circulator

$$[S] = \begin{vmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{vmatrix}$$

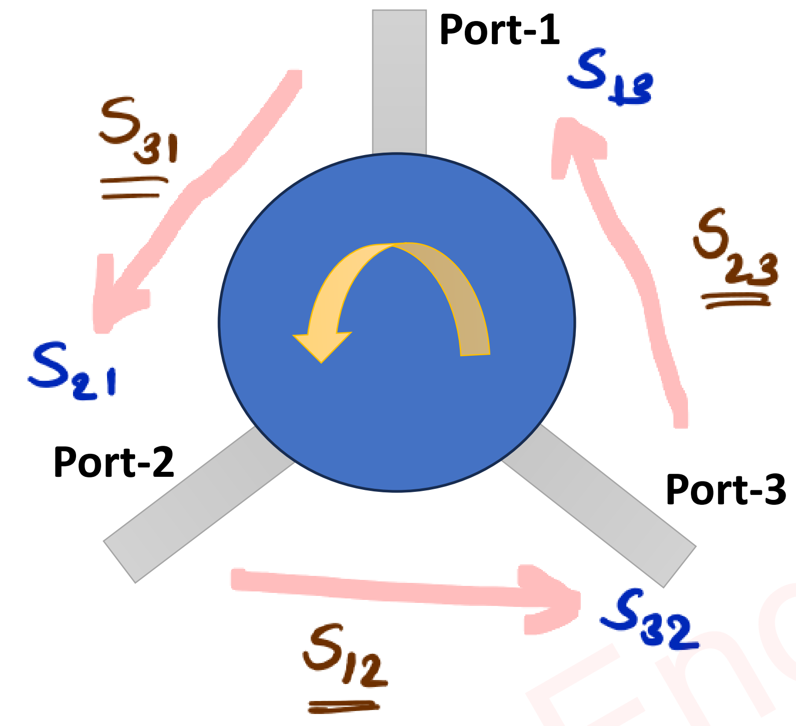
Circulator as Duplexer

- Signal of Transmitter {Port-3} will go Towards Antenna {Port-2} and other ports (Port-1 and Port-4) are Isolated.
- Signals Received by Antenna {Port-2} will go Towards Receiver {Port-1} and other ports (Port-3 and Port-4) are Isolated.
- Here Circulator is used as a Duplexer to Isolate Low Power {Sensitive} Receiver and Higher Power Transmitter.



Example of Circulator

Question 1 – Determine the S Matrix of 3-Port Circulator given insertion loss of 0.5dB, Isolation of 20dB and VSWR of 2.



→ S-parameters for 3-port Circulator

$$[S] = \begin{vmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{vmatrix}$$

→ S_{11}, S_{22} & S_{33} can be Calculated by Reflection Coefficient

$$\Gamma = \frac{VSWR-1}{VSWR+1}$$

$$= \frac{2-1}{2+1}$$

$$= \frac{1}{3} = 0.3333 = S_{11} = S_{22} = S_{33}$$

- S_{12}, S_{23} & S_{31} explain Isolated ports.

$\Rightarrow I = -20\,dB = 20 \log S_{12}$

$\Rightarrow \log S_{12} = -1$

$\Rightarrow S_{12} = 10^{-1} = 0.1 = S_{23} = S_{31}$

- Forward parameters are S_{21}, S_{32} & S_{13}.

$\Rightarrow -0.5 = 20 \log S_{21}$

$\Rightarrow \log S_{21} = -0.025$

$\Rightarrow S_{21} = 10^{-0.025}$

$\Rightarrow S_{21} = 0.944 = S_{32} = S_{13}$

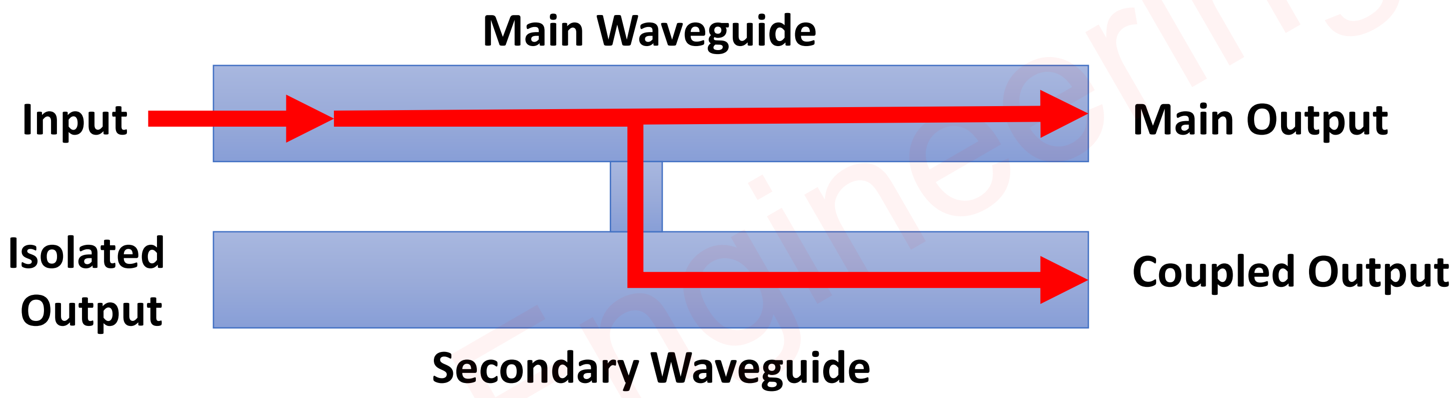
- S-parameters for 3-port circulator

$$[S] = \begin{vmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{vmatrix} = \begin{vmatrix} 0.33 & 0.1 & 0.94 \\ 0.94 & 0.33 & 0.1 \\ 0.1 & 0.94 & 0.33 \end{vmatrix}$$

Directional Coupler

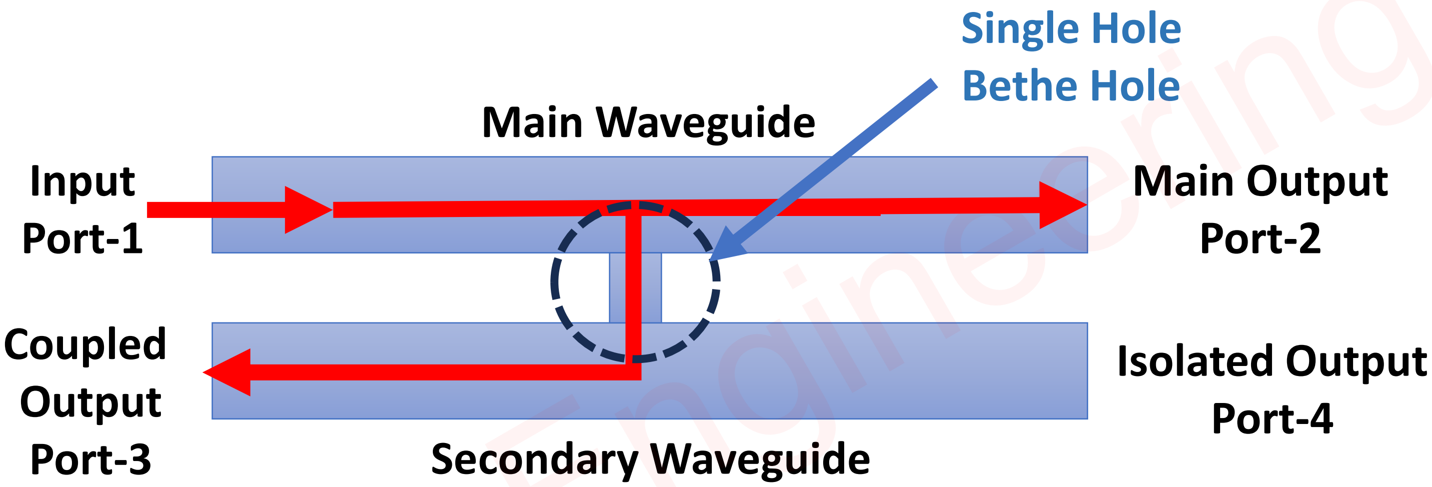
Basics of Directional Coupler

- ❑ Directional Coupler is a 4-Port Device, It is used to Couple Power in Two different Ports.
- ❑ Directional Coupler is designed to sample a specific fraction of power flowing in one Waveguide, delivering it to a secondary line while minimally disturbing the original signal.
- ❑ Directional couplers are fundamental components in many RF applications, including signal sampling, measurement, and routing.



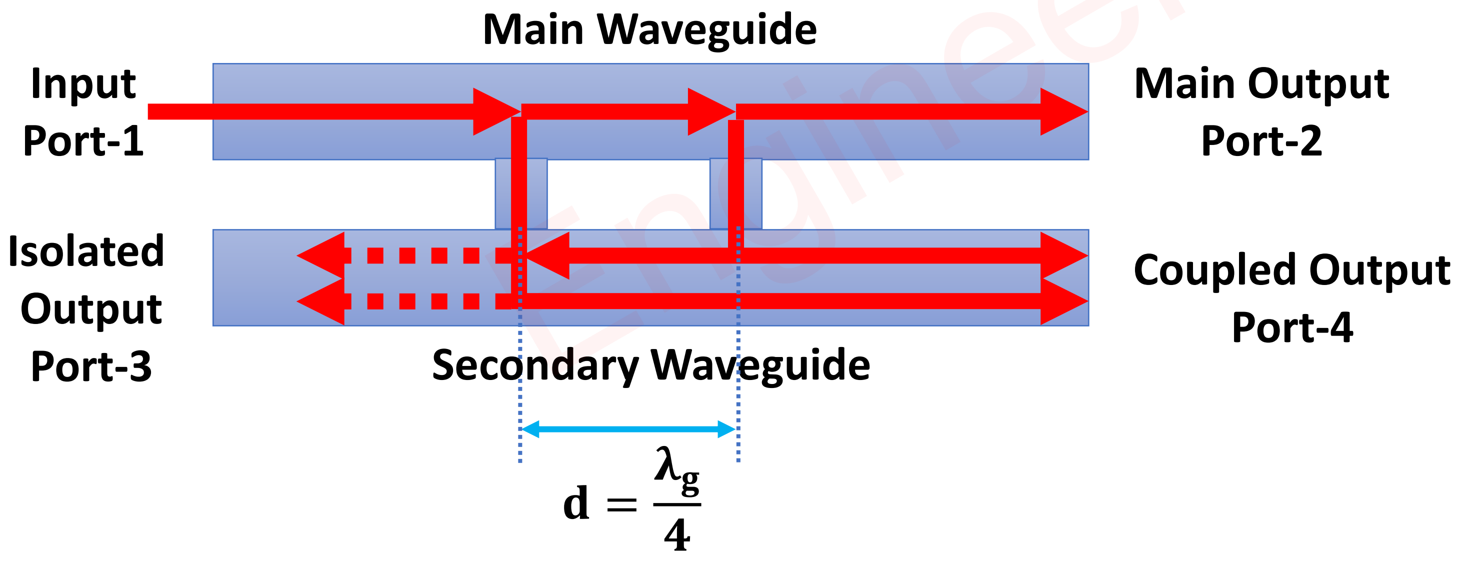
Single Hole Directional Coupler/ Bethe Hole Directional Coupler

- Single Hole Directional Coupler is a 4-Port Device, It is used to Couple Power in Two different Ports (Main Output, Port-2 and Coupled Output, Port-3).
- Power Couple from Main Waveguide to Secondary Waveguide through Single Hole (Bethe Hole).
- Directional Coupler is designed to sample a specific fraction of power flowing in one Waveguide, delivering it to a secondary line while minimally disturbing the original signal.



Two Hole Directional Coupler

- Two Hole Directional Coupler is a 4-Port Device, It is used to Couple Power in Two different Ports (Main Output, Port-2 and Coupled Output, Port-4).
- Power Couple from Main Waveguide to Secondary Waveguide through Two Holes.
- Towards Port-4, the Signal from both holes has Zero phase difference, So It is added at the Coupled Port (Port-4).
- Towards Port-3, the Signal from both holes has a 180^0 phase difference, So It is cancelled at the Isolated Port (Port-3).
- In general spacing between Holes is $(2n+1)\frac{\lambda_g}{4}$.



Parameters of Directional Coupler

❑ **Coupling Factor-C**

$$C = 10\log\left(\frac{P_{In}}{P_{Coupled}}\right)$$

❑ **Directivity-D**

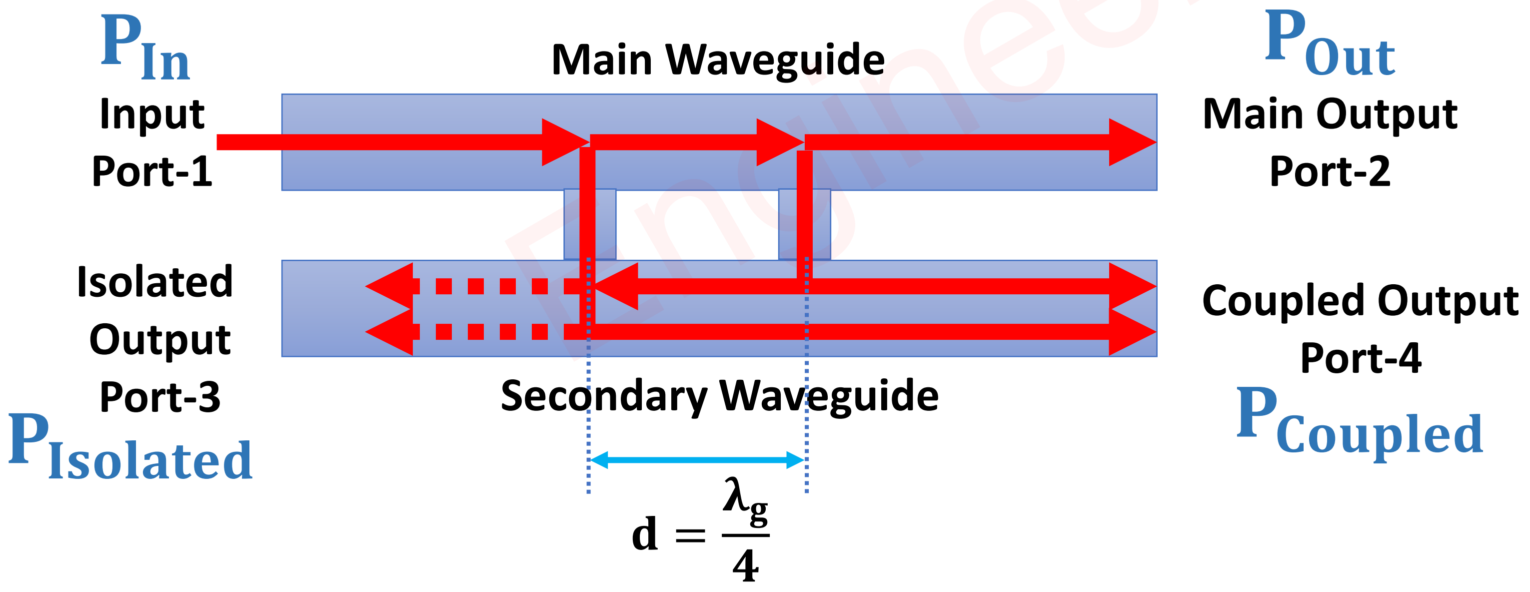
$$D = 10\log\left(\frac{P_{Coupled}}{P_{Isolated}}\right)$$

❑ **Insertion Loss-IL**

$$IL = 10\log\left(\frac{P_{In}}{P_{Out}}\right)$$

❑ **Isolation-I**

$$I = 10\log\left(\frac{P_{In}}{P_{Isolated}}\right)$$

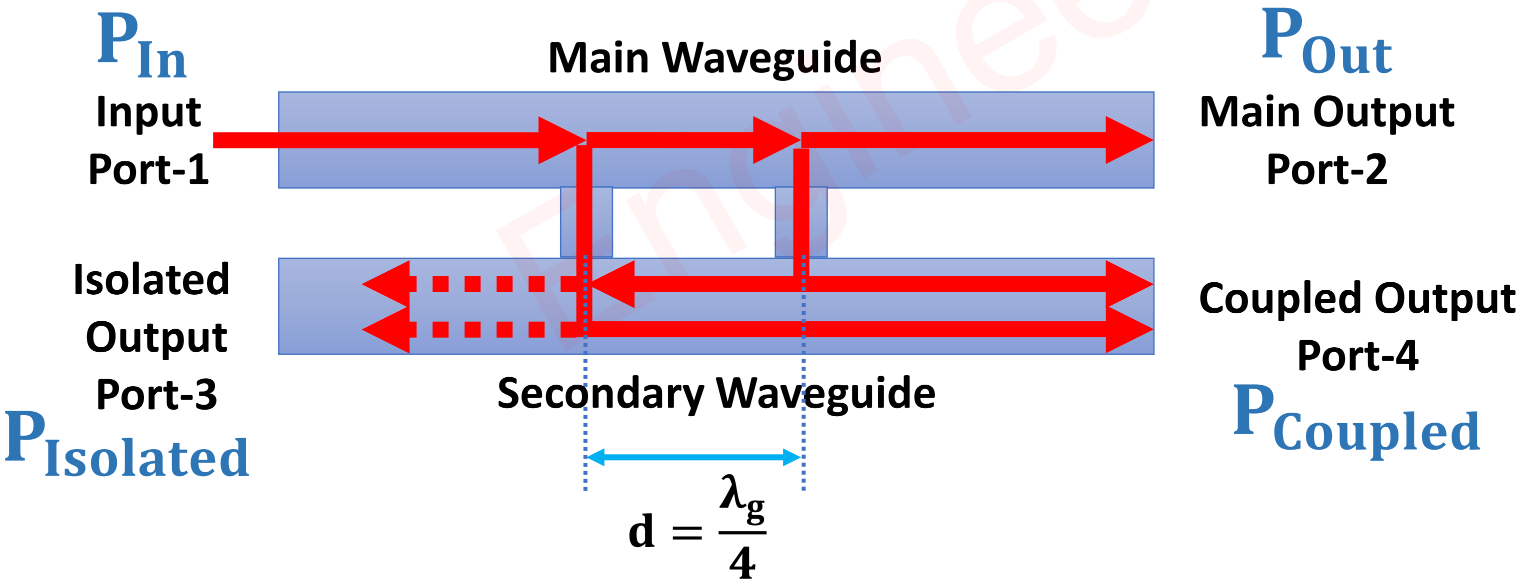


Scattering Parameters of Directional Coupler

❖ Scattering Matrix of 4-Ports Direction Coupler

$$[S] = \begin{vmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{vmatrix} = \begin{vmatrix} 0 & S_{12} & 0 & S_{14} \\ S_{21} & 0 & S_{23} & 0 \\ 0 & S_{32} & 0 & S_{34} \\ S_{41} & 0 & S_{43} & 0 \end{vmatrix} = \begin{vmatrix} 0 & P & 0 & Q \\ P & 0 & Q & 0 \\ 0 & Q & 0 & P \\ Q & 0 & P & 0 \end{vmatrix}$$

- ❑ S_{11}, S_{22}, S_{33} and S_{44} are reflection Coefficients
- ❑ S_{12}, S_{21}, S_{34} and S_{43} are Transmission Coefficients
- ❑ S_{14}, S_{41}, S_{23} and S_{32} are Coupling Coefficients
- ❑ S_{13}, S_{31}, S_{24} and S_{42} are Isolation Coefficients



Applications of Directional Coupler

- **Power Monitoring:**
  - **Sampling a fraction of power for measurement without interrupting the main signal flow.**
- **Signal Routing:**
  - **Directing signals to different paths for processing or measurement.**
- **Isolation and Protection:**
  - **Providing isolation between components to protect sensitive equipment from high-power signals.**
- **Impedance Matching:**
  - **Ensuring proper impedance matching in complex RF networks to minimize reflections.**

Example of Directional Coupler

Question 1 – For a given Directional Coupler if the Input power is 1W, the Forward Power is 0.72W, the Coupled Power is 0.22W, and the Isolated output is 0.01W. Find Ideal S-Matrix.

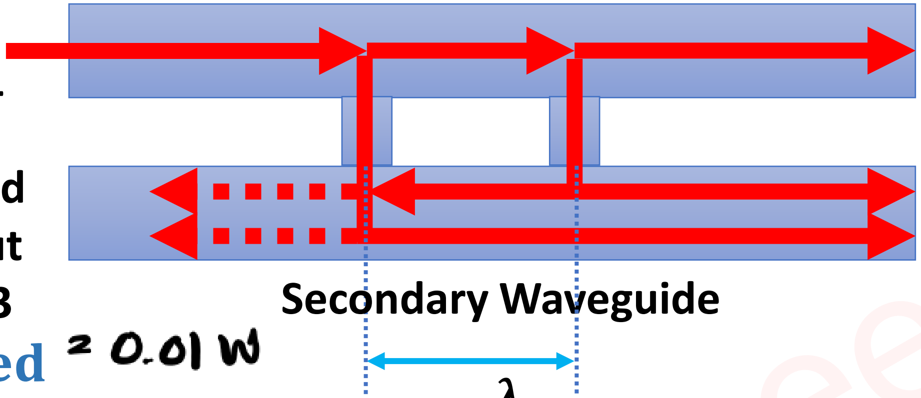
$P_{In} = 1W$

Input
Port-1

Main Waveguide

$P_{Out} = 0.72\ W$

Main Output
Port-2



Isolated
Output
Port-3

$P_{Isolated} = 0.01\ W$

Secondary Waveguide

Coupled Output
Port-4

$P_{Coupled} = 0.22\ W$

$$d = \frac{\lambda_g}{4}$$

$$S_{ij} = \frac{V_{oi}}{V_{ij}} = \sqrt{\frac{P_{oi}}{P_{ij}}}$$

i^{th} o/p Port

j^{th} i/p Port

→ S-Matrix for 4-Port Directional Coupler.

$$[S] = \begin{vmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{vmatrix} = \begin{vmatrix} 0 & 0.84 & 0.1 & 0.46 \\ 0.84 & 0 & 0.46 & 0.1 \\ 0.1 & 0.46 & 0 & 0.84 \\ 0.46 & 0.1 & 0.84 & 0 \end{vmatrix}$$

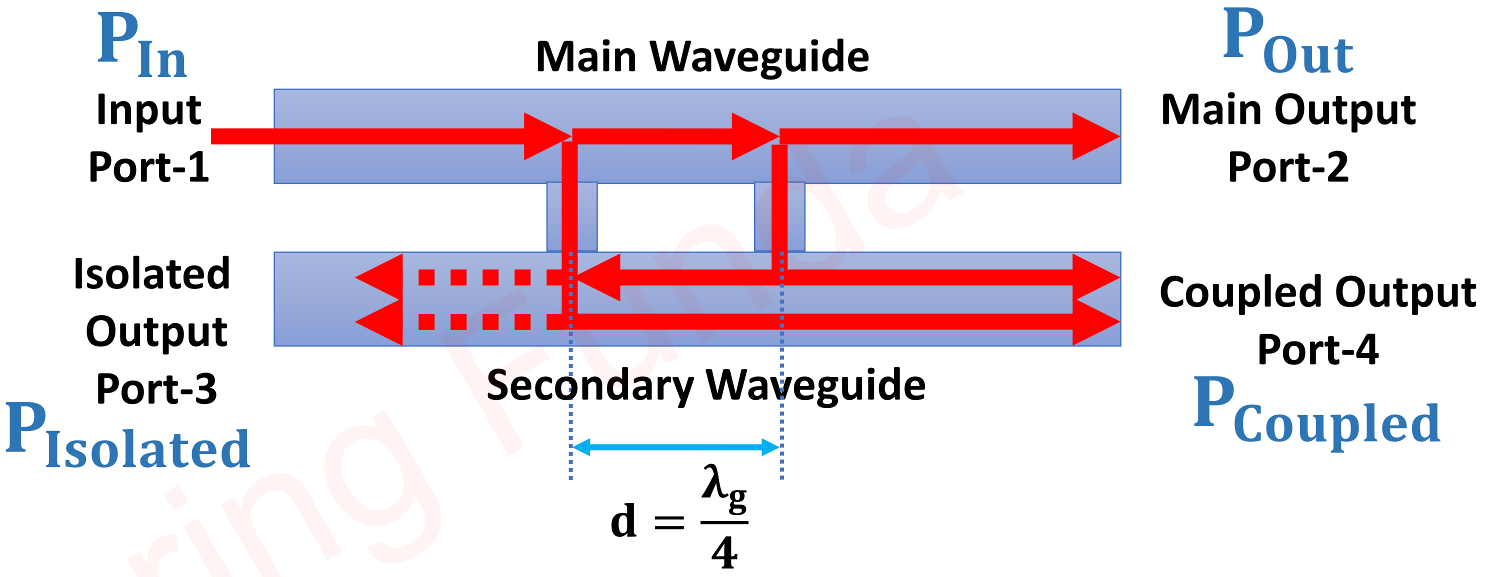
$$\rightsquigarrow \; S_{21} = \frac{V_{o2}}{V_{i1}} = \sqrt{\frac{P_{o2}}{P_{i1}}} = \sqrt{\frac{0.72}{1}} = 0.84 = S_{12} = S_{34} = S_{43}$$

$$\rightsquigarrow \; S_{31} = \frac{V_{o3}}{V_{i1}} = \sqrt{\frac{P_{o3}}{P_{i1}}} = \sqrt{0.01} = 0.1 = S_{13} = S_{24} = S_{42}$$

$$\rightsquigarrow \; S_{41} = \frac{V_{o4}}{V_{i1}} = \sqrt{\frac{P_{o4}}{P_{i1}}} = \sqrt{0.22} = 0.46 = S_{14} = S_{23} = S_{32}$$

Question 2 – Directional Coupler has scattering matrix given below, Find Directivity, Coupling, Isolation, and Return Loss.

$$[S] = \begin{vmatrix} 0.05 & 0.96 & 0.10 & 0.05 \\ 0.96 & 0.05 & 0.05 & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.96 \\ 0.05 & 0.10 & 0.96 & 0.05 \end{vmatrix}$$



→ Return Loss

$$R.L = 20 \log S_{11}$$
$$= 20 \log 0.05$$
$$= -26.02 \text{ dB}$$

→ Coupling

$$C = 20 \log S_{14}$$
$$= 20 \log 0.05$$
$$= -26.02 \text{ dB}$$

→ Isolation I

$$I = 20 \log S_{13}$$
$$= 20 \log 0.1$$
$$= -20 \text{ dB}$$

~ Directivity $D = C - I$

$= -26.02 - (-20\ dB)$

$= -6.02\ dB$

~ $D = 10 \log \left(\frac{P_{coupled}}{P_{iso}} \right)$

$= 10 \log \left(\frac{P_{coupled}}{P_{in}} \times \frac{P_{in}}{P_{iso}} \right)$

$= 10 \log \left(\frac{P_{coupled}}{P_{in}} \right) - 10 \log \left(\frac{P_{iso}}{P_{in}} \right)$

$= C - I$